Financing Firms

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October 29, 2026

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Abstract

Evidence shows that labour and financial markets are closely linked, and shocks are rapidly transmitted between them. Using a model with labour frictions and typical financial frictions, such as costly dividend adjustment and borrowing constraints, this paper shows that the connection between the two markets operates through the firm's marginal value of internal funds. This paper also demonstrates that through the firm's marginal value of internal funds, a low fundamental surplus can amplify shock transmission between the two markets.

1 Introduction

Financial and labour markets are closely connected. Periods of financial distress are typically accompanied by sharp contractions in hiring, declining vacancy creation, and rising unemployment, while improvements in labour market conditions are often

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associated with higher firm valuations and easier access to external finance. These patterns suggest that labour market outcomes and firms' financial decisions are jointly determined. Yet much of the existing literature studies labour market frictions and financial frictions separately, leaving the mechanisms linking the two markets only partially understood.

This paper argues that the connection between labour and financial markets operates through the firm's marginal value of internal funds. In a standard search-and-matching model with frictionless financial markets, an additional unit of internal funds is worth exactly one unit to the firm. Financial conditions therefore play little role in hiring decisions beyond their effect on current profitability. By contrast, when firms face financial frictions, the marginal value of internal funds exceeds unity and varies with the degree of financial stress. Additional internal resources allow firms to reduce the costs associated with dividend adjustment and to relax borrowing constraints. As a consequence, hiring, investment and financing decisions become linked through a common set of shadow values. Financial conditions therefore affect vacancy creation and employment, while labour market developments influence firms' financial choices.

Two financial frictions are required to explain the observed co-movement between labour and financial markets. The first is a cost of adjusting dividends, which makes the marginal value of internal funds depend on the firm's dividends. The second is a borrowing constraint that limits access to external funds. Neither friction on its own generates quantitatively important interactions between labour and financial variables. Instead, strong transmission between the two markets emerges from their interaction. Costly dividend adjustment makes internal funds more valuable, while borrowing constraints make internal funds scarce. Together, they make firms' hiring, investment and financing decisions sensitive to shocks. The nature of the borrowing

constraint is also crucial.

We compare two alternative specifications. The first is a conventional asset-based collateral constraint in which borrowing is limited by the resale value of the firm's capital stock. The second is a cash-flow-based borrowing constraint in which debt capacity is tied to Earnings Before Interest Taxes and Amortisation (EBITDA). While the collateral constraint is widely used in macroeconomic models, the EBITDA-based specification more closely reflects the lending practices faced by many firms in practice. The models show that only the cash-flow-based constraint can generate empirically plausible comovement between labour market and financial-market variables. Under an EBITDA constraint, hiring decisions affect future cash flows and therefore borrowing capacity. Changes in labour market conditions consequently alter firms' financial capacity, creating a powerful feedback mechanism between the two markets. In contrast, under an asset-based constraint, borrowing capacity depends primarily on collateral values, weakening the link between labour market outcomes and financing conditions.

A further contribution of the paper is to identify the fundamental surplus as a key determinant of the strength of these interactions. A lower fundamental surplus reduces the cash flows generated by employment relationships and therefore tightens firms' borrowing constraints. This raises the marginal value of internal funds and amplifies the response of both labour market and financial market variables to shocks. As a result, labour market shocks have larger financial consequences and financial shocks have larger employment consequences when the fundamental surplus is low. The fundamental surplus therefore acts as an amplification mechanism connecting outcomes in the two markets.

To study these mechanisms, we develop a model that combines search frictions in the labour market with costly dividend adjustment and borrowing constraints in the

financial market. The model delivers three main results. First, jointly modelling dividend-adjustment costs and borrowing constraints is essential for generating realistic interactions between labour and financial variables. Second, borrowing constraints based on cash flow substantially outperform conventional collateral constraints in matching the observed co-movement of labour and financial indicators. Third, the sensitivity of both markets to shocks increases as the fundamental surplus declines. Quantitatively, these mechanisms generate a close connection between employment, vacancies, investment and debt that is consistent with observed fluctuations in labour and financial conditions.

Related Literature This paper contributes to the literature that analyses the financial-labour channel through the lens of the Merz (1995); Andolfatto (1996)-style RBC-DMP model. Of this literature Moiseeva (2026) and Garin (2015) are relevant to this paper. Moiseeva (2026) uses a DMP-RBC model with Kiyotaki and Moore (1997) borrowing constraints to explain that the combination of labour and financial frictions results in two-way spillovers between labour and financial markets that amplify shocks. Garin (2015) using a similar model as Moiseeva (2026), also with a Kiyotaki and Moore (1997) borrowing constraint and intra-period bonds in the style of Jermann and Quadrini (2012), identifies similar amplifications following financial and productivity shocks. More generally, Moiseeva (2026) points to the wider consensus in the literature that financial and labour frictions, of various kinds, together amplify aggregate shocks, with which this paper's analysis aligns. The contribution this paper makes to this particular literature is that the combination of EBITDA borrowing constraints, dividend adjustment costs and labour frictions, establish a two-way labour-financial market channel that is amplified by a low fundamental surplus, following both productivity and borrowing constraint shocks.

This paper is also related to the literature on the empirical relevance of earnings-based borrowing constraints. Lian and Ma (2021) shows that, in the data, asset-based lending accounts for only a fifth of US corporate debt by value for medium to large firms, while cash flow-based borrowing constraints govern the rest. Drechsel (2023) shows that, following investment shocks, earnings-based borrowing constraints are empirically more relevant than asset-based borrowing constraints. This paper contributes to the literature by providing theoretical evidence that dividend adjustment costs and EBITDA borrowing constraints amplify shocks. In particular, these financial frictions significantly amplify the effect of borrowing constraint shocks on the labour market. In addition, this paper shows that, under labour and financial frictions, a binding EBITDA borrowing constraint raises the capital-labour ratio in the steady state and can raise the fundamental surplus via a higher marginal product of labour.

Finally, this paper is also related to the literature on the fundamental surplus, which Ljungqvist and Sargent (2017) show that the devices the literature (Shimer, 2005, 2004; Hall, 2005; Hagedorn and Manovskii, 2008) uses to answer the Shimer puzzle all work in the same way: they lower fundamental surplus. This paper contributes to this literature by showing that with EBITDA borrowing constraints and costly dividend adjustment, a low fundamental surplus also amplifies the response of financial and labour market following borrowing constraint shocks, thus it does not only amplify the labour market's response to productivity shocks.

The rest of the paper is structured as follows. Section 2 presents the empirical motivation, and Section 3 presents the model and analytically demonstrates how the interactions between financial and labour frictions establish a labour-financial market transmission channel. Section 4 analyses the role that the financial and labour frictions have on the economy's steady state, while Section 5 analyses their effects on the economy's dynamics.

Section 6 analyses how financial frictions affect the fundamental surplus, and how a low fundamental surplus can amplify the labour-financial market transmission channel. Section 7 concludes.

2 Empirical Motivation

This section presents the empirical motivation. The percentage change in labour market tightness ($\Delta \log \theta_t$) and the net percentage of banks tightening commercial and industrial lending standards ($\Delta \ell_t$), two measures of conditions in the labour and financial markets, move inversely to one another, which is visualised in Figure 1. When the labour market tightens, banks loosen their lending standards, and vice versa. In addition, as Table 1 shows, the two measures are strongly contemporaneously correlated. Taken together, these two indicators suggest an economic link between labour and financial markets, through which shocks are transmitted rapidly. Prior studies have documented spillovers between the two markets. Chodorow-Reich (2014), for example, uses the financial health of firms' lenders at the time of Lehman Brothers' collapse as an exogenous shock to credit supply, and attributes between one-third and one-half of the subsequent employment decline at small and medium-sized firms to the withdrawal of credit. Bell and Machin (2018), using an event study, show that an unanticipated minimum wage hike announcement in the UK on the 8th July, 2015, significantly reduced stock market values for low-wage firms. The evidence, therefore, suggests that labour and financial markets are closely connected and cannot be analysed in isolation.

Despite the close connections between the two markets, the workhorse labour-market model, the Diamond-Mortensen-Pissarides (DMP) model, oversimplifies the financial market. In the model, hiring is treated as an inter-temporal investment. The quantity

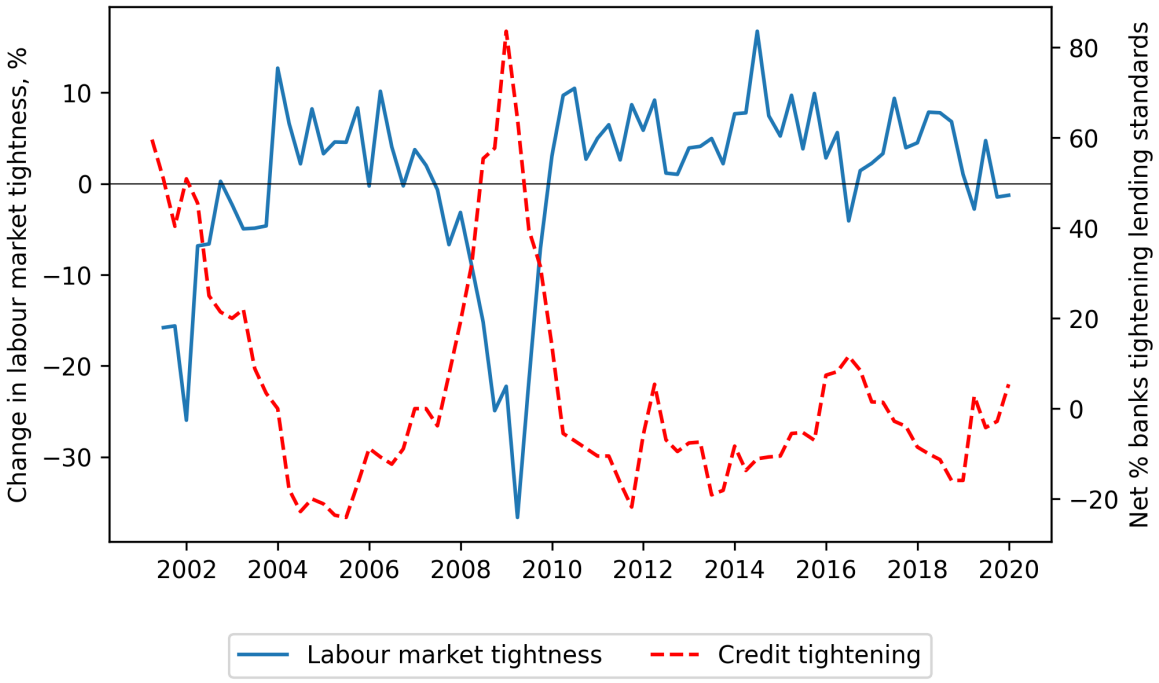


Figure 1: Percentage change in labour market tightness against net percentage credit tightening, 2001Q2 to 2019Q4. Labour market tightness is computed as JOLTS job openings per CPS unemployed worker, while credit tightening is plotted as the net percentage of banks tightening C&I lending standards over the quarter. Source: (U.S. Bureau of Labor Statistics, 2026a,b; Board of Governors of the Federal Reserve System, 2026).

Table 1: Correlation between the change in labour market tightness, $\Delta \log \theta_t$, and the net percentage of banks tightening C&I lending standards i quarters earlier, $\Delta \ell_{t-i}$, 2001Q2 to 2019Q4. Source: (U.S. Bureau of Labor Statistics, 2026a,b; Board of Governors of the Federal Reserve System, 2026).

i	-4	-3	-2	-1	0	1	2	3	4
$\text{Corr}(\Delta \log \theta_t, \Delta \ell_{t-i})$	-0.380	-0.550	-0.710	-0.829	-0.859	-0.827	-0.718	-0.545	-0.325

of hiring, and, in large part, the firm's value of a match, therefore, depends on expected future marginal benefits of workers, which, if large enough, let firms maintain matches under negative cash flow, financing them implicitly by issuing equity costlessly. The drawback of this approach to the labour market is that, first, it fails to capture the costs of issuing equity, which will affect the firm's cash flow and almost certainly its labour demand and the wages it pays. Second, the model ignores firms selling debt, another oversimplification that causes it to overlook how frictions associated with borrowing can affect hiring and wages. Turning to financial market models, particularly those that model returns, like the CAPM, they also tend to overlook the role of search-frictional labour markets. Evidence suggests that labour search frictions strongly determine the cross-section of equity returns (Kuehn, Simutin and Wang, 2017). Intuitively, in frictional labour markets, worker flows are costly to firms and ultimately affect their cash flows.

3 The Model

There is a continuum of firms and households, each of unit mass. Firms are perfectly competitive, identical, and large: they use capital and labour as production inputs, invest in and own their capital, post costly vacancies in a search-frictional labour market, and sell corporate debt. Households consist of a continuum of workers, also of unit mass. Employed workers earn a Nash-bargained wage, and unemployed workers earn a constant level of home production income. The household pools income, perfectly shares unemployment risk from the search-frictional labour market, and chooses consumption and corporate bond purchases.

3.1 Firms

The firm j, at time t, uses capital k_{t-1}^j and labour, n_t^j, to produce output y_t^j, according to,

$$y_t^j = z_t F(k_{t-1}^j, n_t^j) = z_t (k_{t-1}^j)^\alpha (n_t^j)^{1-\alpha} \tag{1}$$

where z_t is a stochastic economy-wide neutral productivity and F is the Cobb-Douglas production function. Capital depreciates at rate δ, so investment is $k_t^j - (1-\delta)k_{t-1}^j$.

The firm posts vacancies v_t^j at constant cost $\gamma > 0$ and loses a constant fraction $s \in (0,1)$ of workers at a constant separation rate. Let q_t denote the probability that a vacancy is filled, so the law of motion of labour is,

$$n_{t+1}^j = (1-s)n_t^j + q_t v_t^j \tag{2}$$

Aggregate matches follow the Cobb-Douglas matching function,

$$H_t = \Upsilon V_t^{1-\eta} U_t^\eta \tag{3}$$

where V_t is aggregate vacancies, U_t aggregate unemployment, Υ is the matching efficiency and η is the elasticity of matches with respect to unemployment. The labour market's tightness, vacancy filling rate and the job finding rate therefore are,

$$\theta_t = \frac{V_t}{U_t}, \qquad q_t = \frac{H_t}{V_t} = \Upsilon \theta^{-\eta}, \qquad f_t = \frac{H_t}{U_t} = \Upsilon \theta^{1-\eta} \tag{4}$$

with $q_t, f_t \in (0,1)$.

The firm sells risk-free one-period bonds x_t^j to households and receives $1/\tilde{R}_t$ per bond to fund investment and smooth dividends. Following Jermann and Quadrini (2012) the firm receives a tax-benefit on the interest they pay on debt. $\tilde{R}_t$ is the effective gross interest rate that the firm pays on its debt, and the tax benefit is $\tau > 0$, such that,

$$\tilde{R}_t = 1 + (R_t - 1)(1 - \tau)$$

where $R_t = 1 + r_t$ is the gross interest rate households receive. Since,

$$\Delta_t = \frac{x_t}{\tilde{R}_t} - \frac{x_t}{R_t} > 0$$

the government transfers, $T_t = -\Delta_t$, to households to pay for the tax benefit.

The firm's borrowing is constrained by a multiple its EBITDA,

$$\frac{x_t^j}{\tilde{R}_t} \leq \xi_t \left(y_t^j - w_t^j n_t^j - \gamma v_t^j \right) \tag{5}$$

where w_t^j is the wage. The term in brackets $(y_t^j - w_t^j n_t^j - \gamma v_t^j)$ is the accounting definition of the firm's EBITDA. $\xi_t > 0$ is the EBITDA multiple, which can be interpreted as the number of periods of current earnings a lender will advance. In this model, a borrowing constraint shock is a shock to ξ. A tightening shock, for instance, is one that reduces ξ. From (5), a growing labour force increases the firm's borrowing capacity if and only if the marginal product of labour exceeds the marginal cost. An important characteristic of (5) is that it makes the firm's borrowing capacity flow-based rather than stock-based,

like a Kytatki-Moore (KM) borrowing constraint, which will be used as a comparison,

$$\frac{x_t^j}{\tilde{R}_t} \leq \xi_t^{KM}(1-\delta)k_t^j \tag{6}$$

in which the firm's borrowing is collaterised against its undepreciated capital in the following period. ξ_t^{KM} is the fraction of the undepreciated capital that is used as collateral[1]. A firm that is borrowing constrained by (5) can quickly adjust its borrowing capacity by adjusting the vacancies it posts. A firm borrowing constrained by (6), conversely, will see its borrowing capacity adjust more slowly following shocks.

Households hold constant and equal shares in every firm, receive profits as dividends, and instruct firms to maximise the present value of the lifetime stream of profits. Following Jermann and Quadrini (2012), firm j pays dividends d_t^j and bears a quadratic cost when it deviates from the long-run target ϕ. The cost of paying dividends follows,

$$\psi(d_t^j) = d_t^j + \kappa\left(d_t^j - \phi\right)^2 \tag{7}$$

where $\kappa \geq 0$ is the adjustment cost parameter, which can be interpreted as the cost of substituting between raising finance with debt and reducing dividends. Alternatively, κ can capture managers' preference for smoothing dividends, which Jermann and Quadrini indicate has been confirmed in the literature and can arise from agency problems. In this case, the larger the κ, the stronger the managers' preference for smoothing dividends. κ has no effect on a steady state when ϕ is set equal to the steady state dividends.

[1]In steady state, ξ^{KM} is set to one.

The firm's flow budget, therefore, is,

$$y_t^j - w_t^j n_t^j - \gamma v_t^j + \frac{x_t^j}{\tilde{R}_t} = x_{t-1}^j + k_t^j - (1-\delta)k_{t-1}^j + \psi(d_t^j) \tag{8}$$

Using (7) and (8), it can be shown that when the borrowing constraint binds, if the firm's expenditures exceed its income, it must reduce its dividends. Suppose the required reduction in dividends is from ϕ to $\phi - \Delta_d$, where $\Delta_d > 0$; then, given the adjustment cost, the resources that become available to the firm following the reduction are

$$\Delta_d - \kappa \Delta_d^2$$

As κ increases, the maximum resources the firm can obtain through dividend reductions decreases, and the firm must cut more dividends to achieve the same desired increase in available resources. If $\kappa = 0$, then the firm can obtain any desired increase in resources by cutting dividends by the same amount, and even raise equity via negative dividends.

3.2 The Firm's Problem, Recursively

The aggregate state is,

$$S \equiv (z, \xi, N, K, X)$$

where N is aggregate employment, K is aggregate capital, and X is aggregate debt, which evolves according to $S' = \Gamma(S, z', \xi')$ The firms invididual state is,

$$\omega \equiv (n, k, x)$$

The firm's optimisation problem is,

$$J(\omega; S) = \max_{d,v,k',x'} \{d + \mathbb{E}\left[m' J(\omega'; S')\right]\} \tag{9}$$

where m' is the owner's stochastic discount factor, subject to the flow budget

$$y - wn - \gamma v + \frac{x'}{\tilde{R}} = x + k' - (1-\delta)k + \psi(d)$$

the law of motion of labour (2), and the EBITDA borrowing constraint

$$\frac{x'}{\tilde{R}} \leq \xi \left(y - wn - \gamma v\right)$$

Remark 1. *With $\tau > 0$, the firm will find it profitable to finance itself with debt over equity; thus, the borrowing constraint will bind. Combining the binding borrowing constraint with the budget yields,*

$$1 + \xi = \frac{x + k' - (1-\delta)k + \psi(d)}{y - wn - \gamma v} > 1 \tag{10}$$

(10) shows that, when the borrowing constraint binds, ξ determines the percentage difference between firms' debt repayments, investment, and dividends and their EBITDA.

Using these constraints, the Bellman may be written in the form,

$$J(\omega; S) = \max_{d,v,k',x'} \Bigg\{ d + \mathbb{E}\left[m' J(\omega'; S')\right] + \mu \left[y - wn - \gamma v + \frac{x'}{\tilde{R}} - x - k' + (1-\delta)k - \psi(d) \right] + \Lambda \left[\xi(y - wn - \gamma v) - \frac{x'}{\tilde{R}} \right] \Bigg\}$$

μ is the shadow price of the budget constraint, and Λ is the shadow price of the borrowing constraint. From the two constraints, increases in output raise the firm's value function

through the budget and borrowing constraints in the current period. Similarly, labour and hiring costs lower the firm's value function through both constraints in the current period. Variables not in EBITDA, such as dividends, debt and capital investment, affect the firm's value differently. Dividends, for instance, only affect the firm's value through tightening the budget constraint. In the current period, capital investment reduces the firm's value by tightening the budget constraint, but in the following period, it raises the firm's value through more resources being available for dividend payments and a looser borrowing constraint. Debt sales, on the other hand, raise the firm's value by providing resources today; this is offset by a tighter borrowing constraint today and fewer internal funds available in the future, which reduce the firm's value.

3.3 The Labour–Finance Transmission Mechanism Through the Firm's Optimality Conditions

This section examines the labour-financial market link in the EBITDA economy through the firm's optimality conditions, and compares them to the optimality conditions in the KM economy. For the full derivations of the following optimality conditions in the EBITDA economy, see Appendix A. For the KM economy's optimality conditions, see Appendix F.

Dividends. The marginal benefit of dividends to the firm owner is unity, while the marginal cost consists of the costs of paying the dividend, $\psi'(d)$, priced at μ. The optimality condition for dividends is, therefore,

$$1 = \mu\psi'(d) \tag{11}$$

where $\psi'(d) = 1 + 2\kappa(d - \phi)$. Let $\mu \equiv 1 + \hat{\mu}$, which means (11) becomes,

$$1 + \hat{\mu} = \frac{1}{1 + 2\kappa(d - \phi)}$$

after rearranging,

$$\hat{\mu} = \frac{-2\kappa(d - \phi)}{1 + 2\kappa(d - \phi)} \tag{12}$$

$\hat{\mu}$ shows the budget constraint's bite compared to the case where $\kappa = 0$. If dividends are at their target, $d = \phi$, then the budget constraint's bite is the same as in the $\kappa = 0$ case. When dividends are below target, $d < \phi$, then $\hat{\mu} > 0$, as the firm is in financial distress, while if dividends are above target, the budget constraint binds less than in the $\kappa = 0$ case.

Internal Funds The marginal value of internal funds for the firm under the EBITDA borrowing constraint depends on both shadow prices μ and Λ. If we define internal funds, Φ, as,

$$\Phi \equiv y - wn - \gamma v - x \tag{13}$$

the marginal value of internal funds under the EBITDA borrowing constraint is,

$$\frac{\partial J(\omega; S)}{\partial \Phi} = 1 + \hat{\mu} + \xi\Lambda \tag{14}$$

In the absence of financial frictions, $\hat{\mu} = \Lambda = 0$, the marginal value of internal funds is equal to unity. With financial frictions and the binding borrowing constraint, this rises above unity since additional internal funds let firms bring dividends to their target and loosen the borrowing constraint. If the firm's borrowing constraint follows (6), on the

other hand, the marginal value of internal funds is,

$$\frac{\partial J(\omega; S)}{\partial \Phi} = 1 + \hat{\mu}$$

Under an asset-based borrowing constraint, therefore, the marginal value of internal funds for the firm only depends on whether the firm meets its dividend target.

Vacancies. The cost of posting a vacancy for the firm includes the unit cost γ, which reduces the firm's value by $\gamma(1 + \hat{\mu})$ units, and the $\xi\gamma$ loss of borrowing capacity, valued at $\xi\gamma\Lambda$, resulting from a tighter borrowing constraint. Consequently, the marginal cost of posting vacancies is $\gamma(1 + \hat{\mu} + \xi\Lambda)$, the product of the unit cost of posting a vacancy and the marginal value of internal funds, while the marginal benefit is the discounted value of an additional worker, given the probability q that the vacancy is filled. At the optimum, the following condition must hold,

$$q\mathbb{E}\left[m' J_n(\omega'; S')\right] = \gamma(1 + \hat{\mu} + \xi\Lambda) \tag{15}$$

This condition is an extended version of the standard DMP free-entry condition. If dividends are below target or the borrowing constraint bites harder, the higher marginal value of internal funds raises the marginal cost of hiring, and the firm posts fewer vacancies. When dividends are at their target, or $\kappa = 0$, and the firm has no tax advantage on debt (i.e., $\tau = 0$ and $\Lambda = 0$), the borrowing constraint does not bind, and the condition reduces to the standard DMP free-entry condition.

Compare (15) to the vacancy optimality condition if the firm's borrowing constraint is

(6),

$$q\mathbb{E}\left[m' J_n(\omega'; S')\right] = \gamma(1 + \hat{\mu})$$

While the marginal cost of hiring is determined by the marginal value of internal funds as well, it is only affected by how much the firm's dividend stray from their target, and there is no direct link to the borrowing constraint's tightness. The way in which the KM borrowing constraint affects the firm's hiring is via the changes in the future expected marginal product of capital on the future expected marginal product of labour. Alternatively, if changes in the firm's capital investment change $\hat{\mu}$.

Workers. An additional worker in the current period raises the firm's output by the marginal product of labour, less the wage, which the firm values according to,

$$(1 + \hat{\mu} + \xi\Lambda)\left[zF_n(k, n) - w\right]$$

This means that the firm's marginal value of internal funds determines how much a firm values an additional worker. If, for instance, the marginal value of internal funds is high, it will value a match with a worker more. The value of an additional worker can be obtained from the envelope condition with respect to n, and thus obeys,

$$J_n(\omega; S) = (1 + \hat{\mu} + \xi\Lambda)\left[zF_n(k, n) - w\right] + (1 - s)\mathbb{E}\left[m' J_n(\omega'; S')\right] \tag{16}$$

This is analogous to the value of a filled vacancy in the DMP model.

If the firm were to be KM borrowing constrained, the envelope condition would be,

$$J_n(\omega; S) = (1 + \hat{\mu})\left[zF_n(k, n) - w\right] + (1 - s)\mathbb{E}\left[m' J_n(\omega'; S')\right]$$

As before, the marginal value of internal funds determines how much a firm values a match, however, this is not directly linked to the firm's borrowing constraint.

Capital. The marginal cost of purchasing an additional unit of capital is $1+\hat{\mu}$. The marginal benefit of capital investment is the expected future marginal product of capital, valued at the expected future marginal value of internal funds, plus the undepreciated remaining unit the firm will have, valued at $1+\hat{\mu}'$. At the optimum,

$$\mathbb{E}\Big[m'\Big\{(1+\hat{\mu}'+\xi'\Lambda')z'F_k(k',n')+(1+\hat{\mu}')(1-\delta)\Big\}\Big]=1+\hat{\mu} \tag{17}$$

which is the firm's capital Euler equation. The expected future marginal value of internal funds changes capital demand. If the firm expects a rise in the marginal value of internal funds, the marginal benefit of capital will rise, and the firm will invest more.

Remark 2. *When dividends are at their target, and the borrowing constraint bites, $\Lambda>0$, the capital-labour ratio will be larger than when $\hat{\mu}=\Lambda=0$. Using the Cobb-Douglas production function, and letting ρ be the discount rate, the capital-labour ratio in the steady state is,*

$$\frac{k}{n}=\left[\left(\frac{1+\hat{\mu}+\xi\Lambda}{1+\hat{\mu}}\right)\frac{\alpha z}{\rho+\delta}\right]^{\frac{1}{1-\alpha}} \tag{18}$$

When the borrowing constraint binds, and $\hat{\mu}=0$, capital investment raises the firm's borrowing capacity and the firm's desired steady state capital-labour ratio. When the borrowing constraint does not bind, and $\hat{\mu}=0$, then (18) reduces to the well-known neoclassical growth model capital-labour ratio. (18) also shows that due to financial frictions raising the marginal value of internal funds, the steady state ratio is larger than in the financially frictionless case.

If the firm were to be constrained by the KM borrowing constraint, capital's optimality condition would be instead,

$$\mathbb{E}\left[m'\Big\{(1+\hat{\mu}')\left[z'F_k(k',n')+1-\delta\right]\Big\}\right]+\Lambda\xi^{KM}(1-\delta)=1+\hat{\mu}$$

In contrast to (17), the marginal benefit of capital arises in the current period from loosening the borrowing constraint today and from the additional expected future funds generated by the newly installed capital in the following period. As a result, the firm's investments are less forward-looking than under EBITDA borrowing constraints. For example, in the EBITDA borrowing constraint case, if the firm were to expect a larger Λ in the future, it would increase its investments in anticipation. In the KM economy, the expected future tightness of the borrowing constraint does not change the firm's behaviour.

Debt. A unit of borrowing delivers $1/\tilde{R}$ of resources to the budget, adding $(1+\hat{\mu})/\tilde{R}$ units of value, and consumes $1/\tilde{R}$ of borrowing capacity, worth $\Lambda/\tilde{R}$ units of value. The optimality condition is,

$$\frac{\Lambda}{\tilde{R}}+\mathbb{E}\left[m'(1+\hat{\mu}')\right]=\frac{1+\hat{\mu}}{\tilde{R}} \tag{19}$$

which is the same as in the KM borrowing constraint case. (19) is the firm's debt Euler equation. The tax advantage associated with debt financing ensures that $\tilde{R}<R$ and $\Lambda>0$. In the steady state, where $m'=1/R$ and $\hat{\mu}'=\hat{\mu}=0$, then,

$$\Lambda=1-\frac{\tilde{R}}{R}$$

If $\tau=0$, then $\tilde{R}=R$, and $\Lambda=0$, in which case the firm's equity-debt choice would be indeterminant.

3.4 Households

A continuum of identical households, each of unit mass, holds a continuum of workers, also of unit mass, of which n are employed and $1 - n$ are unemployed. Employed workers earn a Nash-bargained wage w, and unemployed workers receive a constant home-production income b. Workers separate at rate s and find jobs at rate f, but they pool their income inside the household, so unemployment risk is perfectly shared. Households own constant and equal shares in every firm, earning the dividend d, and buy a' units of the bonds firms sell at the price $1/R$. The government funds the firms' interest deduction with a lump-sum tax levied uniformly on households,

$$T = \frac{X'}{R} - \frac{X'}{\tilde{R}},$$

where X' is the aggregate quantity of bonds traded.

Households have CRRA preferences over consumption, c,

$$u(c) = \frac{c^{1-\sigma}}{1-\sigma}$$

where σ is the CRRA parameter. $W(a, n; S)$ is the household's value function, hence its optimisation problem is,

$$W(a, n; S) = \max_{c, a'} \left\{ u(c) + \beta \mathbb{E} \left[W(a', n'; S') \right] \right\}, \tag{20}$$

subject to the budget

$$c + \frac{a'}{R} = wn + (1-n)b + a + d + T \tag{21}$$

and the law of motion of labour,

$$n' = (1-s)n + f(1-n) \tag{22}$$

Bond holdings satisfy the Euler equation,

$$u'(c) = \beta R \, \mathbb{E}\left[u'(c')\right] \tag{23}$$

which prices the one-period risk-free bond and makes the gross real rate R move with the consumption path. Using this Euler equation, the firm's discount factor is, $m' = \beta u'(c')/u'(c)$.

3.5 Nash Wages

The wage is determined through Nash bargaining between the worker and the firm,

$$w = \arg\max_{w} \left\{ \left(W_n(a, n; S)\right)^{\Psi} J_n(\omega; S)^{1-\Psi} \right\} \tag{24}$$

where $W_n(a,n) = \partial W(a,n)/\partial n$ is the household's marginal value of employment and Ψ is the worker's bargaining power. The resulting frictional Nash wage is[2]

$$w = \Psi\left[zF_n(k,n) + \frac{(1-s)\gamma}{q}\left[1 - \mathbb{E}\left[\frac{1+\hat{\mu}+\xi\Lambda}{1+\hat{\mu}'+\xi'\Lambda'}\right]\right] + \gamma\theta\mathbb{E}\left[\frac{1+\hat{\mu}+\xi\Lambda}{1+\hat{\mu}'+\xi'\Lambda'}\right]\right] + (1-\Psi)b \tag{25}$$

where, in the steady state when $\hat{\mu} = \hat{\mu}'$, $\xi' = \xi$ and $\Lambda = \Lambda' > 0$, or in the absence of frictions, which is when $\hat{\mu} = \Lambda = 0$, the wage simplifies to,

$$w^f = \Psi\left[zF_n(k,n) + \theta\gamma\right] + (1-\Psi)b$$

where w^f stands for the frictionless wage. Unlike Monacelli, Quadrini and Trigari (2023), debt does not enter the wage because it does not affect the value of an additional worker to the firm or the value of an additional employed worker to the household, and therefore is not part of the joint surplus. Financial frictions, $\hat{\mu}$ and Λ, enter the wage, however. To see the effect of the financial frictions on the wage, subtract the frictionless wage from (25), and letting $\Omega \equiv \mathbb{E}\left[\frac{1+\hat{\mu}+\xi\Lambda}{1+\hat{\mu}'+\xi'\Lambda'}\right]$, which yields,

$$w - w^f = \Psi\frac{\gamma}{q}\left[(1-s) - f\right](1-\Omega) \gtreqless 0$$

In the absence of financial frictions, $\Omega = 1$, and then $w - w^f = 0$. If $(1-s) > f$, then a match is more likely to remain in each period than an unemployed worker is likely to find a match. If $\Omega < 1$, then the frictional wage rises over the frictionless wage. In this case, the future marginal value of internal funds rises, and the firm begins valuing the incumbent workforce more because they generate output and support its borrowing capacity without the firm having to undertake costly hiring. Conversely, if $\Omega > 1$, then the frictional wage falls below the frictionless wage.

[2]See Appendix B for the derivation.

Turning to the case where $(1-s) < f$, the effects reverse. New matches are easier to create, so changes in financial conditions affect the value the firm assigns to its incumbent workforce in the opposite way.

In the case of the KM borrowing constraint, the wage is

$$w = \Psi \left[zF_n(k,n) + \frac{(1-s)\gamma}{q} \left[1 - \mathbb{E}\left[\frac{1+\hat{\mu}}{1+\hat{\mu}'} \right] \right] + \gamma\theta\mathbb{E}\left[\frac{1+\hat{\mu}}{1+\hat{\mu}'} \right] \right] + (1-\Psi)b$$

As expected, the wage in the KM economy is only indirectly affected by the borrowing constraint, via the changes in the marginal value of internal funds under the KM borrowing constraint, $1+\hat{\mu}$, or changes to the marginal product of labour.

Definition 3.1 (Recursive Competitive Equilibrium)**.** A recursive competitive equilibrium in the EBITDA economy consists of the set of household's value function $W(a,n;S)$ and its policy functions $c(a,n;S)$, $a'(a,n;S)$, the firm's value function $J(\omega;S)$ and its policy functions $d(\omega;S)$, $v(\omega;S)$, $k'(\omega;S)$, $x'(\omega;S)$, aggregate prices $R(S)$, $w(S)$, and $m(S,S')$, labour market tightness $\theta(S)$ and the law of motion of aggregate states $S' = \Gamma(S)$, such that,

1. **Firm Optimisation.** Given $R(S)$, $w(S)$, and $\theta(S)$, and the aggregate law of motion Γ, the firm's policies are optimal and $J(\omega;S)$ solves the firm's Bellman equation.

2. **Household Optimisation** Given $R(S)$, $w(S)$, and $\theta(S)$, the household's policies are optimal and $W(a,n;S)$ solves the household's Bellman equation.

3. **Wage Determination.** The wage $w(S)$ solves the Nash bargaining problem (24).

4. **Aggregation and Matching.** By symmetry, $n = N$, $k = K$ and $x = X$. This means in the aggregate, $Y(S) = zF(K, N)$, $D(S) = d(N, K, X; S)$, $V(S) = v(N, K, X; S)$, $K'(S) = k'(N, K, X; S)$, $U(S) = 1 - N$, $X'(S) = x'(N, K, X; S)$. Where $C(S)$ is aggregate consumption, the stochastic discount factor for firms is $m(S, S') = \beta u'(C(S'))/u'(C(S))$. Matches satsify $H(S) = \Upsilon V(S)^{1-\eta} U(S)^{\eta}$, thus $\theta(S) = V(S)/U(S)$, $q(S) = \Upsilon\theta(S)^{-\eta}$, and $f(S) = \Upsilon\theta(S)^{1-\eta}$. Labour evolves according to $N'(S) = (1 - s)N + q(S)V(S)$.

5. **Taxes and Market Clearing.** Households purchase firms' debt, thus, $a'(X, N; S) = X'(S)$ is the bond market clearing. This means $C(S) = c(X, N; S)$. Taxes satisfy,

$$T(S) = \frac{X'(S)}{R(S)} - \frac{X'(S)}{\tilde{R}(S)}$$

and the goods market clears according to,

$$Y(S) + (1 - N)b - \gamma V(S) = C(S) + K'(S) - (1 - \delta)K + \kappa(D(S) - \phi)^2$$

6. **Aggregate Law of Motion.** The aggregate law of motion $S' = \Gamma(S, z', \xi')$ is consistent with agents' decisions and z and ξ's stochastic processes.

4 Labour-Financial Market Link in the Steady State

This section analyses the role that the interactions between the financial frictions, specifically the borrowing constraints, and labour market frictions have on an economy's steady state.

4.1 Parametrising the EBITDA economy's steady state

The vacancy cost is calibrated to 0.11 and matching efficiency to 0.31, and the separation rate is s is set to 0.02, targeting a steady state unemployment rate of $U = 6\%$, and a steady state labour market tightness of $\theta = 1$. Home production income is set to $b = 0.7$, and δ is set to 0.107 to ensure the dividend steady state remains non-negative while bringing the steady state marginal product of labour close to unity.

β is set to 0.985 and α is set to 0.36 following Jermann and Quadrini (2012). Following Fujita and Ramey (2012) the matching elasticity is set to $\eta = 0.7$ and a worker bargaining weight is $\Psi = 0.7$. The constant risk aversion parameter σ is set to 2 and the steady state productivity z is set to 1. Following evidence from Lian and Ma (2021), the steady state EBITDA multiple ξ is set to 9.12, meaning that, with each period representing a quarter, firms, in the steady state, can borrow up to 2.3 times their annual earnings. Turning to the tax τ, rounding up the top statutory rate a US corporation faces once state taxes are added to the federal rate, which was 39.1% in 2012 (Congressional Budget Office, 2017), it is set to 0.4. This value is higher than Jermann and Quadrini's τ of 0.35 because the authors consider solely federal taxes. The dividend target ϕ is set to the dividend steady state.

4.2 Parametrising the Alternative Economies

The EBITDA economy's steady state is compared to two alternatives in the following analysis. The first is an otherwise identical frictionless baseline where $\tau = 0$ and firms

face no borrowing constraint. In this economy, similar to the RBC model, firms borrow to invest in new capital. In this frictionless baseline, there are still DMP labour frictions because removing these would change the interpretation of labour to hours worked, making the model incomparable to the KM and EBITDA economies. The second alternative is a version where the borrowing constraint is asset-based and follows (6). The parameters γ and Υ are calibrated targeting the same unemployment rate and labour market tightness, set ϕ to the steady state level of dividends, and keep the remaining parameters at the same values as in the EBITDA economy. $\xi^{KM} = 1$ in the steady state.

4.3 Comparing the Steady States

Table 2 compares the steady states of the three economies.

Table 2: Steady states across the three economies, 2 d.p.

		Frictionless baseline	EBITDA Economy	KM Economy
Y	Output	1.73	1.78	1.77
zF_n	Marginal product of labour	1.18	1.21	1.21
w	Wage	1.16	1.19	1.18
U	Unemployment	0.06	0.06	0.06
θ	Tightness	1.00	1.00	1.00
Λ	Collateral multiplier	–	0.01	0.01
μ	Firm budget multiplier	–	1.00	1.00
X	Debt	5.08	5.97	4.92
K	Capital	5.08	5.52	5.46
d	Steady State Dividends	0.01	0.00	0.02

The EBITDA borrowing constraint affects the steady state through the capital-labour ratio, as shown in (18). The binding borrowing constraint raises the marginal product of labour through a larger K/N; the EBITDA economy's marginal product of labour is 1.211, while the KM economy's is 1.206. This raises the EBITDA economy's output and Nash wages, but because the larger capital stock raises total production costs, its dividends fall to 0.004.

5 Labour-Financial Market Link in the Dynamics

This section analyses the labour-financial transmission channel in EBITDA, KM and baseline economies, via impulse response shocks to z, ξ and ξ^{KM}. The shock processes in this experiment for z, ξ and ξ^{KM} are,

$$
\begin{aligned}
z_t &= (1-\rho_z)z_{ss} + \rho_z z_{t-1} + \varepsilon_t^z, \\
\xi_t &= (1-\rho_\xi)\xi_{ss} + \rho_\xi \xi_{t-1} + \varepsilon_t^\xi, \\
\xi_t^{KM} &= (1-\rho_{\xi^{KM}})\xi_{ss}^{KM} + \rho_{\xi^{KM}}\xi_{t-1}^{KM} + \varepsilon_t^{\xi^{KM}}.
\end{aligned}
$$

The persistence are set such that $\rho_\xi = \rho_z = \rho_{\xi^{KM}} = 0.95$, and $\sigma_z = \sigma_\xi = \sigma_{\xi^{KM}} = 0.01$. For the shocks, $\varepsilon_t^z \sim N(0, (z_{ss}\sigma_z)^2)$, $\varepsilon_t^\xi \sim N(0, (\xi_{ss}\sigma_\xi)^2)$ and $\varepsilon_t^{\xi^{KM}} \sim N(0, (\xi_{ss}^{KM}\sigma_{\xi^{KM}})^2)$.

To identify the effects of interactions between borrowing constraints and labour frictions on dynamics, for each shock, the responses of the three economies are shown under $\kappa = 0$. To show the effects of dividend adjustment costs, borrowing constraints, and labour frictions on dynamics, the responses of the three economies under $\kappa = 0.03$ are presented as well. Treating Λ as the counterpart to the credit tightening measure in Figure 1, this non-negative κ is chosen so that the model's implied ratio of the standard deviation of $\Delta \log \theta$ to the standard deviation of $\Delta \log \Lambda$ is aligned with the ratio computed from the data presented, which is 0.4. Table 3 shows the implied standard deviations from the EBITDA and KM economies across different κ values for the productivity and borrowing constraint shocks.

Table 3: Relative volatility across values of κ

		Shock	
Model	κ	Productivity	Borrowing Constraint
EBITDA	0.05	0.178	0.048
EBITDA	0.03	0.229	0.052
EBITDA	0.01	0.409	0.061
EBITDA	0.00	0.181	0.058
KM	0.05	0.279	0.007
KM	0.03	0.326	0.007
KM	0.01	0.246	0.007
KM	0.00	0.204	0.037

Note: The entries report the $\text{sd}(\Delta \log \theta)/\text{sd}(\Delta \log \Lambda)$ computed from the impulse responses.

The results show that $\kappa = 0.01$ brings the EBITDA economy closest to target, while $\kappa = 0.03$ does for the KM economy, each following a productivity shock. Under borrowing constraint shocks, the large response of Λ, which increases with κ, brings each model's implied volatility ratio below the target. The comparative analysis therefore uses $\kappa = 0.03$ as the shared value because, at this level, both economies underestimate the standard deviation ratio by the most similar amount. Nevertheless, this paper's analysis and conclusions do not depend on this particular choice of κ.

5.1 Productivity Shock

The way a productivity shock is transmitted from production, through the labour market and then the financial market depends on the response of the marginal value of internal funds.

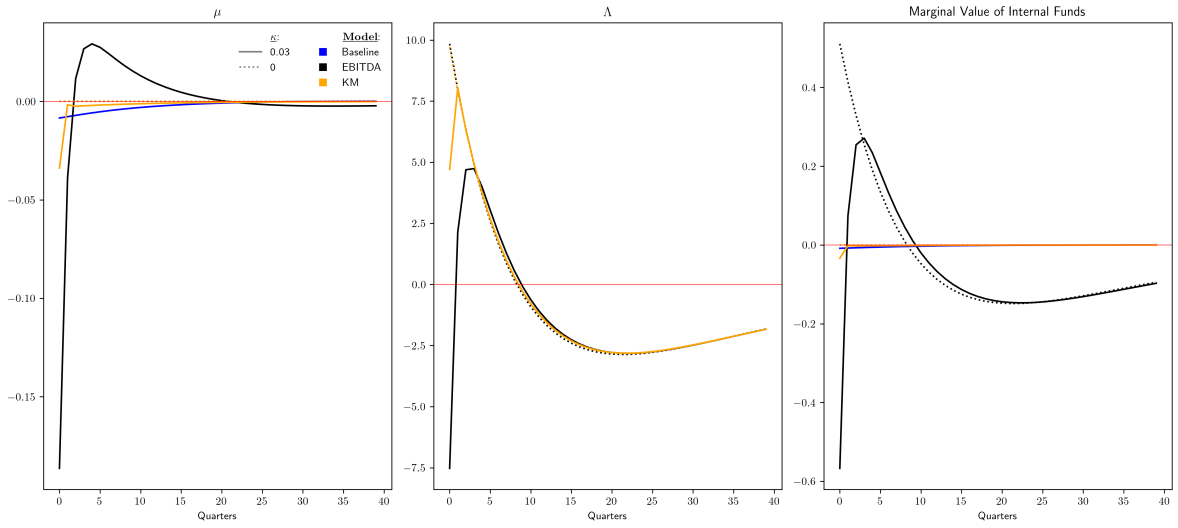


Figure 2: Percentage deviations from the steady state, 1% productivity shock.

Figure 2 compares the marginal value of internal funds across the three economies and the changes in the shadow prices μ and Λ. Beginning with the EBITDA economy with costly dividend adjustment, the productivity shock initially reduces the marginal value of internal funds. This differs from the case with no dividend adjustment costs. In that case, initially following the shock, the marginal value of internal funds rises. The difference stems from how frictions change the firm's financing of investment and hiring over the shock. With dividend adjustment, to stay close to the target, the firm relies less on selling debt because it uses more of its additional revenue to invest and hire rather than pay dividends. In this case, the shadow prices μ and Λ fall. Without dividend adjustment costs, however, the firm relies more on borrowing to finance investment

and hiring, since it pays no penalty for paying as high a dividend as possible. This raises the shadow price Λ. Turning to the baseline and KM economies, the marginal value of internal funds does not change much. In both economies, the tightness of the borrowing constraint does not affect the marginal value of internal fund, therefore, the marginal value of internal funds does not change over the shock. Considering Λ in the KM economy, this rises initially over the shock because the firm does not immediately experience an increase in its borrowing capacity.

Figures 3 and 4 illustrate the effects of different paths of the marginal value of internal funds on labour- and financial-market variables. Both figures show that the path of the marginal value of internal funds, induced by a specific set of financial frictions, changes the shape of the economy's response to the shock. We see this in the difference between the responses in the KM- and EBITDA-borrowing-constrained economies. The figures also show that dividend adjustment costs are required for shocks to be rapidly transmitted across labour and financial markets in the EBITDA economy. This can be seen by comparing the black dotted lines to the solid black lines. In the KM and baseline economies, however, due to their weak impact on the marginal value of internal funds, dividend adjustment costs have little impact on the response of labour and financial variables following the productivity shock. This result is consistent with the mechanisms identified from the firm's optimality conditions. Large changes in the marginal value of internal funds strongly affect the firm's hiring demand, which spills over into its borrowing capacity and investment decisions. In the KM economy, however, although the borrowing constraint tightens significantly after the shock, its disconnect from the firm's marginal value of internal funds yields minimal differences between the dividend adjustment cost and $\kappa = 0$ cases.

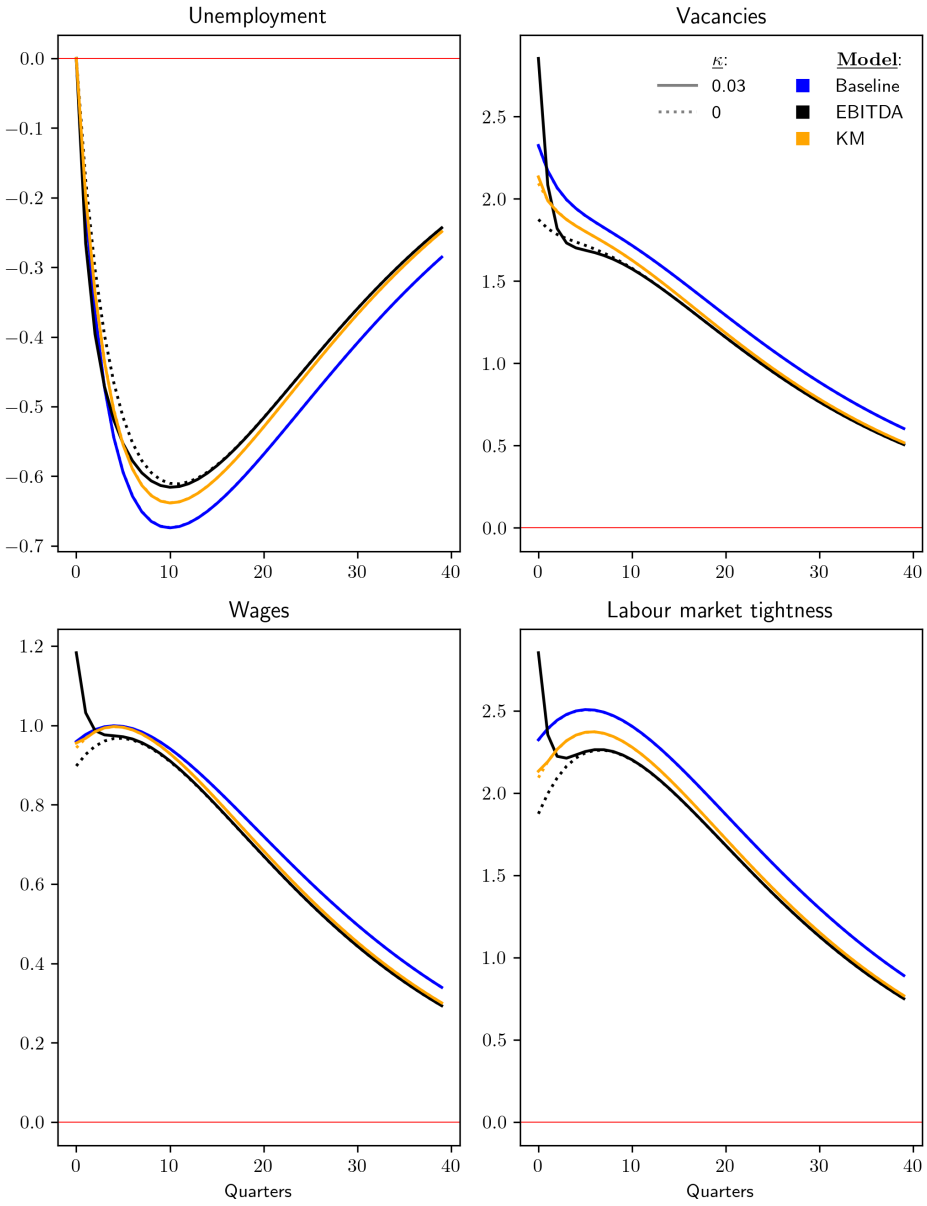


Figure 3: Percentage deviations from the steady state, 1% productivity shock.

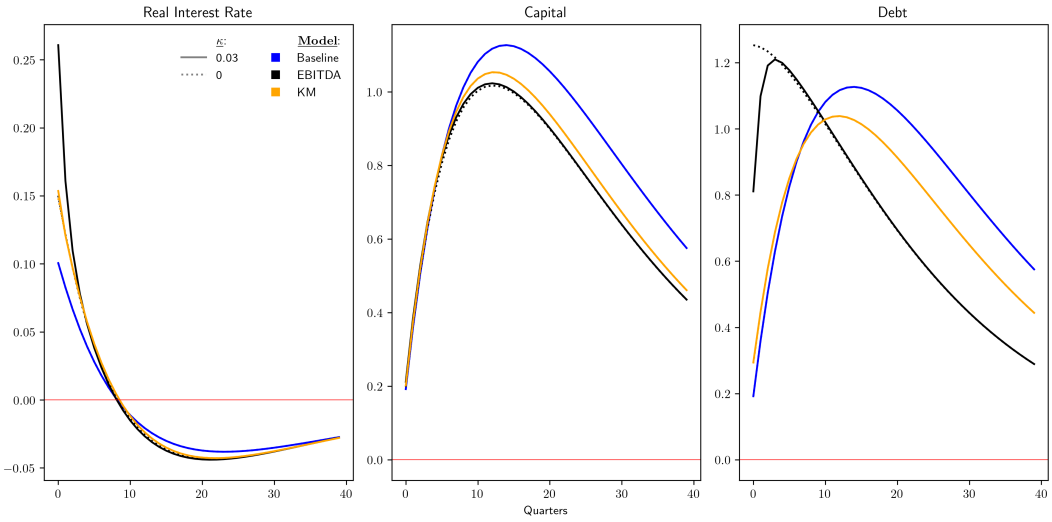


Figure 4: Percentage deviations from the steady state, 1% productivity shock.

The rapid transmission of the shock to the labour market in the EBITDA economy with $\kappa = 0.03$, driven by a sharp decline in the marginal value of internal funds, can be seen via the amplification of labour market tightness and wages. For instance, compared with the EBITDA economy with $\kappa = 0$, labour market tightness and wage responses are 30% to 50% larger with dividend adjustment costs. Similarly, compared with the baseline and KM economies, these variables respond by 20 percentage points to 35 percentage points more in the EBITDA economy with dividend adjustment costs. Intuitively, the sharp transmission reflects the shock lowering the marginal value of internal funds and the marginal cost of posting vacancies by an equally sharp amount. In the alternative economies, however, the marginal value of internal funds does not change as much in response to the shock, or, as in the EBITDA economy with $\kappa = 0$, it may even rise. This results in a lesser decline in the marginal cost of vacancies, a lesser demand for labour upon the shock's impact,

Turning to the financial market, the productivity shock immediately increases the EBITDA-constrained firm's borrowing capacity, resulting in more debt sold and a

larger fall in its price. In the KM economy, the small change in the marginal value of internal funds, combined with the slow expansion of the firm's borrowing capacity as its capital stock builds up, does not lead to as large an increase in debt sales. The KM borrowing-constrained firm's capital response rises above the EBITDA-constrained economy's capital response, as the former needs more capital to expand its borrowing capacity than the latter to take advantage of the productivity shock.

5.2 Borrowing Constraint Shock

Turning to consider the transmission of a shock to the borrowing constraint across financial and labour markets, the analysis is similar to before.

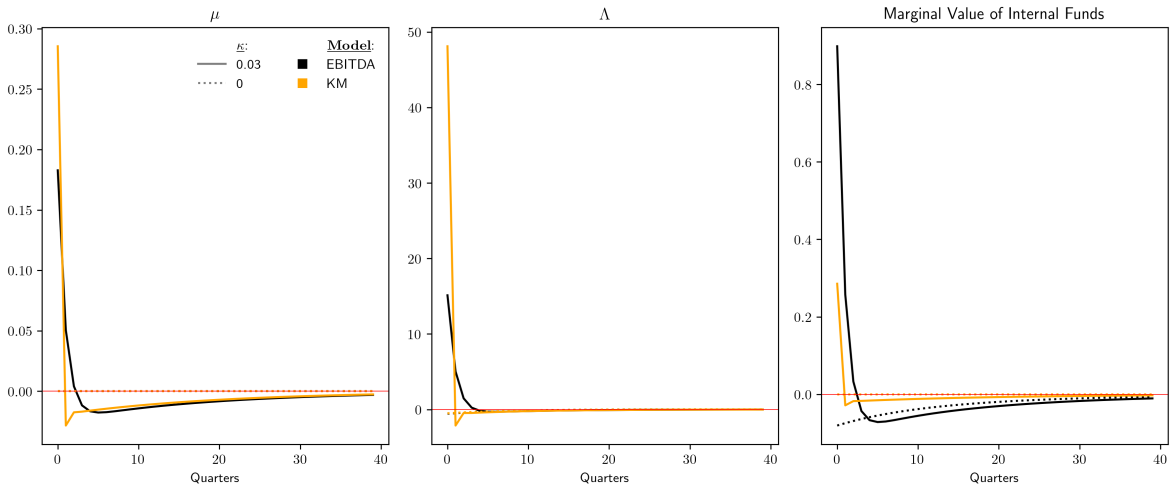


Figure 5: Percentage deviations from the steady state, −1% borrowing constraint shock.

Figure 5 illustrates the importance of dividend adjustments for the marginal value of the internal fund's response to a shock to the borrowing constraint. Beginning with the EBITDA economy, with $\kappa = 0$, the marginal value of internal funds, despite a borrowing constraint shock, falls as the firm frictionlessly cuts dividends to reduce its reliance on borrowing to finance hiring and investment. With dividend adjustment,

however, as expected, the tightening of the borrowing constraint raises the marginal value of internal funds. The firm can no longer cut dividends costlessly to rely less on borrowing; therefore, as credit becomes scarcer, it values its internal funds more. In the KM economy, as before, the marginal value of internal funds equals the shadow price μ. This means that, without dividend adjustment, the firm's marginal value of internal funds does not change. With dividend adjustment costs, the marginal value of internal funds rises only when dividends fall below target. These differences in the marginal value of internal funds dictate responses in the labour and financial markets, as Figures 6 and 7 show.

The strength of the shock's transmission onto the labour market depends on the size of the change in the marginal value of internal funds. The EBITDA borrowing-constrained economy with dividend adjustment costs sees the largest rise in the marginal value of internal funds; thus, vacancies, wages, and labour market tightness fall the most. In the absence of borrowing constraints, however, the shock has the opposite, unexpected effect: it loosens the borrowing constraint and lowers the marginal value of internal funds in the EBITDA economy.

Remark 3. *To see why under $\kappa = 0$ the shadow price Λ falls, we can set $\hat{\mu} = \hat{\mu}' = 0$ in the firm's debt Euler equation, to obtain,*

$$\Lambda = 1 - \frac{1 + (R-1)(1-\tau)}{R}$$

where for $R > 0$, $\partial\Lambda/\partial R > 0$, which is intuitive: the larger the real interest rate that households receive on debt, the tighter the borrowing constraint. Since the gross rate evolves according to the Euler equation,

$$R = \frac{1}{\beta} \frac{u'(c)}{\mathbb{E}\left[u'(c')\right]}$$

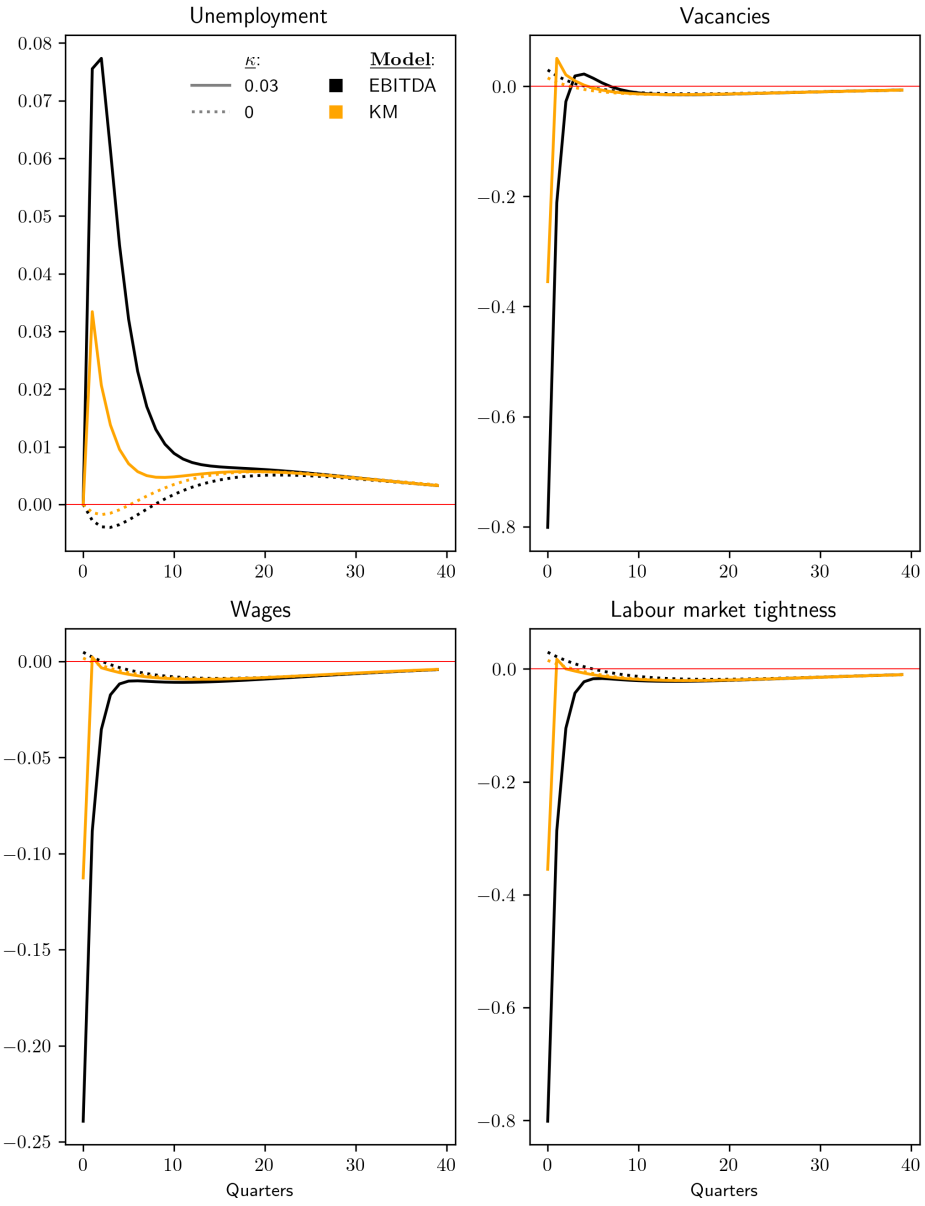


Figure 6: Percentage deviation from the steady state, −1% borrowing constraint shock.

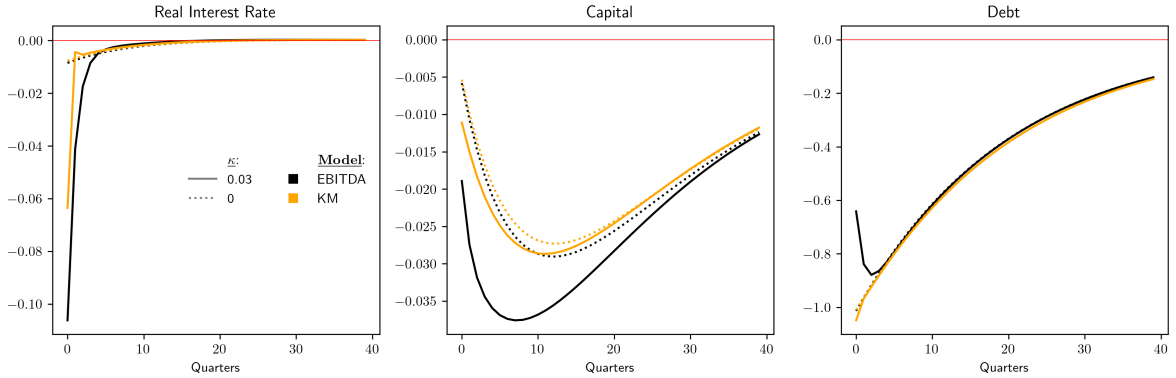


Figure 7: Percentage deviation from the steady state, −1% borrowing constraint shock.

The mechanism that reduces Λ *under the borrowing-constraint shock is the expected future path of declining consumption, which increases demand for bonds as workers smooth consumption and reduces the real interest rate workers receive and firms pay, thereby reducing the bite of the firm's borrowing constraint.*

The marginal value of internal funds also affects the financial market. The larger fall in labour, due to the larger rise in the marginal value of internal funds, reduces capital's marginal product, investment and real interest rates; however, due to the EBITDA borrowing constraint, the responses of μ, Λ and debt are the largest in the KM economy. This is because, after the shock, firms in the KM economy must reduce investment and borrowing. Their capacity to mitigate the drop in borrowing is weaker than under EBITDA constraints. Firms under EBITDA borrowing constraints can cut vacancies to maintain some cash flow during the shock, so the decline in bond sales is smaller.

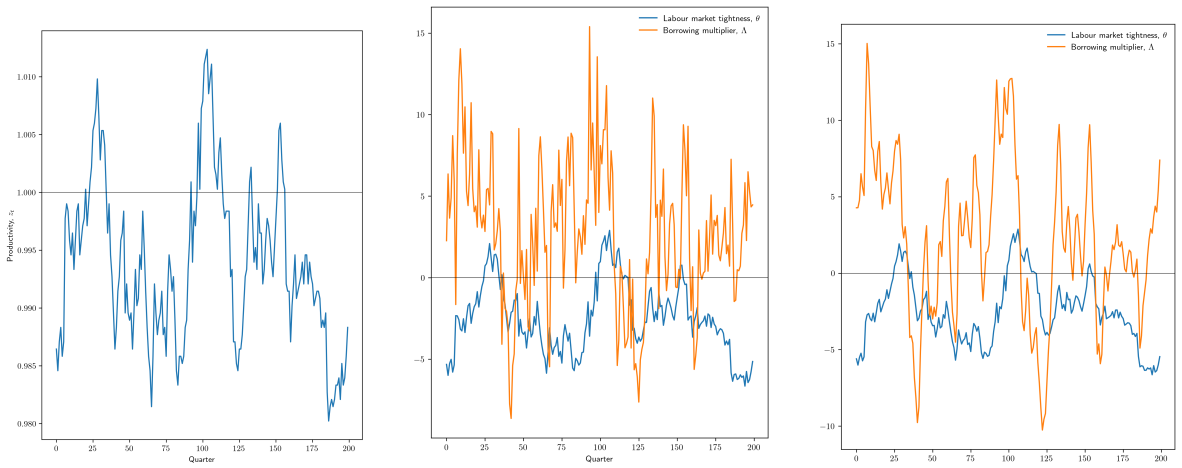
5.3 Qualitative Labour–Finance Comovement in Simulated Cycles

This section compares the qualitative consistency of the labour-finance comovement in the data with the comovement generated by the EBITDA and KM economies. In particular, the following analysis tests whether the EBITDA and KM economies reproduce the sign and timing of the relation between labour-market and financial conditions documented in Figure 1 over a simulated sequence of productivity and borrowing-constraint shocks. The analysis again uses Λ as the model's counterpart to the credit tightness measure presented earlier.

As before, the EBITDA and KM economies are set with $\kappa = 0.03$, and the productivity and borrowing constraint shocks are simulated separately. To compute the simulations, the productivity and borrowing constraint shocks are discretised, with the same persistence as before, and an unconditional standard deviation of 1% in logs, into a first-order Markov chain using the Rouwenhorst (1995) method. The simulated sequences of z, ξ and ξ^{KM}, over a 10,000-period time horizon, are fed into the model's first-order impulse responses to construct the implied paths of θ and Λ.

Productivity Shock Simulation Figure 8 presents the productivity-driven fluctuations.

Following the sequence of productivity shocks, both economies show a negative relation between θ and Λ. Considering $\Delta \log \theta$ and $\Delta \log \Lambda$, like in Figure 1, the EBITDA economy produces a contemporaneous correlation of -0.72, while the KM economy generates 0.70. The EBITDA economy, following the simulated productivity shocks,

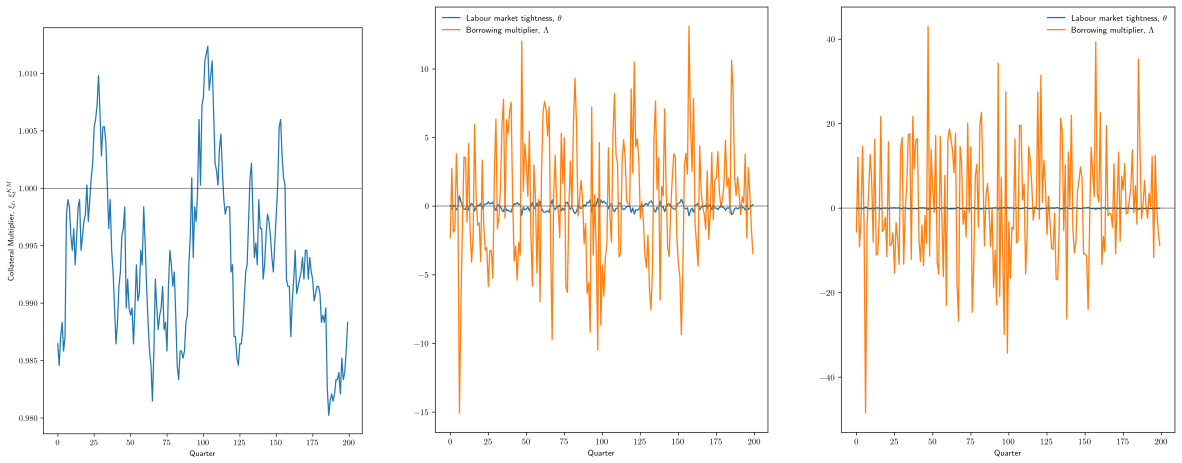


(a) z Path (b) EBITDA economy (c) KM economy

Figure 8: Percentage deviations from the steady states over z's simulated path.

reproduces the sign can be seen from the data.

Borrowing Constraint Shock Simulation Figure 9 plots the simulated path of ξ and ξ^{KM} and the corresponding movements in θ and Λ.



(a) ξ and ξ^{KM}'s Path (b) EBITDA economy (c) KM economy

Figure 9: Percentage deviations from the steady states over ξ and ξ^{KM}'s simulated path.

As the impulse responses show, Λ is more volatile than θ in the simulations. Both economies generate a contemporaneous correlation of -1 between $\Delta \log \theta$ and $\Delta \log \Lambda$.

This is expected since a change in ξ or ξ^{KM} immediately changes a firm's borrowing capacity, inducing an immediate change in how much the borrowing constraint bites and in the firm's capacity to post vacancies.

Correlations: Leads and Lags Table 4 presents lead-lag correlations between $\Delta \log \theta$ and $\Delta \log \Lambda$, computed from the simulations. Each row shows the correlations from each model for the productivity and borrowing constraint shocks.

Table 4: Correlation between $\Delta \log \theta_t$, and $\Delta \log \Lambda$, i quarters earlier.

$\text{Corr}(\Delta \log \theta_t, \Delta \log \Lambda_{t-i})$, i	-4	-3	-2	-1	0	1	2	3	4
EBITDA, productivity	-0.049	0.024	0.184	0.712	-0.720	0.075	0.017	0.019	0.008
KM, productivity	-0.200	-0.223	-0.273	0.474	0.699	0.031	0.039	0.029	0.025
EBITDA, collateral	0.013	0.052	0.107	0.335	-1.000	0.321	0.102	0.050	0.012
KM, collateral	-0.008	0.012	-0.029	0.520	-1.000	0.531	-0.031	0.011	-0.009

While the EBITDA and KM economies do not reproduce the lead-lag pattern observed in Table 1, under productivity shocks, it is only the EBITDA economy that reproduces a contemporaneous correlation between $\Delta \log \theta$ and $\Delta \log \Lambda$ that is similar to the data. Under borrowing constraint shocks, by contrast, both economies consistently predict a negative contemporaneous correlation between the two measures. Borrowing constraints, therfore, allow the model to produce a contemporaneous correlation between changes in labour and credit tightness measures that is observed in the data following borrowing constraint shocks. Following productivity shocks, however, the EBITDA economy with dividend adjustment costs qualitatively aligns more closely with the data. As seen from the impulse responses, the reason is that under the EBITDA borrowing constraint and dividend adjustment costs, positive productivity shocks loosen the firm's borrowing constraint, while symmetrically tighten under negative productivity shock.

6 The Fundamental Surplus Matters

This section examines how the fundamental surplus, the quantity of resources that the invisible hand can allocate to vacancy creation (Ljungqvist and Sargent, 2017), amplifies the labour-finance transmission channel in the EBITDA borrowing-constrained economy following productivity and borrowing constraint shocks.

6.1 The Fundamental Surplus under the EBITDA Borrowing Constraint in the Steady State

Before proceeding to examine the productivity and borrowing constraint impulse responses, under different fundamental surpluses, let us consider the impact of the financial frictions on the fundamental surplus.

Taking the marginal product of capital in the steady state,

$$zF_k(k,n) = z\alpha k^{\alpha-1}n^{1-\alpha} = \frac{\beta^{-1} - (1-\delta)}{1 + \xi\frac{\Lambda}{1+\hat{\mu}}}$$

the production function's decomposition,

$$y = n \cdot zF_n(k,n) + k \cdot zF_k(k,n)$$

and the Ljungqvist and Sargent (2017) fundamental surplus definition,

$$\text{FS} = zF_n(k,n) - b$$

the fundamental surplus can be written as,

$$\text{FS} = z^{\frac{1}{1-\alpha}} \left[\frac{1 + \xi \frac{\Lambda}{1+\hat{\mu}}}{\beta^{-1} - (1-\delta)} \right]^{\frac{\alpha}{1-\alpha}} \alpha^{\frac{\alpha}{1-\alpha}} (1-\alpha) - b \tag{26}$$

A binding borrowing constraint, $\Lambda > 0$, ceteris paribus, raises the marginal product of labour through a higher capital-to-labour ratio, and thus raises the fundamental surplus. This is similar to the case where $\hat{\mu} < 0$ in the steady state: a larger capital-labour ratio raises the fundamental surplus through the same channel.

In the steady state, a low fundamental surplus reduces the firm's borrowing capacity. Using the steady state wage, the EBITDA borrowing constraint, the production function's decomposition into its marginal products, the free-entry condition and the matching function, the firm's bond sales can be written as,

$$\frac{x}{\widetilde{R}} = \xi \frac{\Upsilon \theta^{1-\eta}}{s + \Upsilon \theta^{1-\eta}} \left[\frac{\alpha}{1-\alpha} (\text{FS} + b) + \frac{\gamma \rho}{\Upsilon} \theta^{\eta} \right] \tag{27}$$

(27) shows that when the fundamental surplus is low, the firm has lower EBITDA cash flow, and it can borrow less.

6.2 Quantitative Analysis

Let us now consider the effect of the fundamental surplus on the labour and financial markets following productivity and borrowing constraint shocks, in the EBITDA borrowing constraint economy. The following analysis focuses on two versions of the model: a low fundamental surplus version where b is 0.9, and a high fundamental surplus version where b is 0.4. For the rest of the parameters in both models, the same calibration

strategy is used as before. The same steady state unemployment and labour market tightness targets remain the same, ϕ is also set to equal the respective steady state dividends, and the remaining parameters remain the same as before.

Regarding the effect that dividend adjustment costs have on the dynamics, the following analysis also considers the cases of $\kappa = 0$ and $\kappa = 0.1$ for both high and low fundamental surplus economies. Setting $\kappa = 0.1$ illustrates how a large adjustment cost changes the dynamics across different fundamental surpluses.

6.2.1 Productivity Shock

Beginning with a positive productivity shock, Figure 10 shows the goods market, labour and financial market responses under the two fundamental surpluses.

As expected from Ljungqvist and Sargent (2017), θ's increase following the positive productivity shock is larger when the fundamental surplus is low. In fact, between the $b = 0.9$ and $b = 0.4$ economies, the θ increase in the former is more than twice larger than in the latter. In addition, under a lower fundamental surplus there is a slight increase in output and investment responses to the shock, and greater amplification in changes in Λ and μ. The low fundamental surplus, under dividend adjustment costs, causes a larger decline in Λ upon impact by approximately 80%, and an approximately 30% larger decline in μ upon impact, and thus, as shown in Figure 11, a larger decline in marginal value of internal funds. The dividend adjustment costs, via the marginal value of internal funds, as before, increase the shock's transmission onto labour market tightness and wages. This can be seen for both $b = 0.4$ and $b = 0.9$ cases. In the absence of borrowing constraints, thus with a smaller positive response of the marginal value of internal funds, we observe that debt responds minimally to a low fundamental surplus.

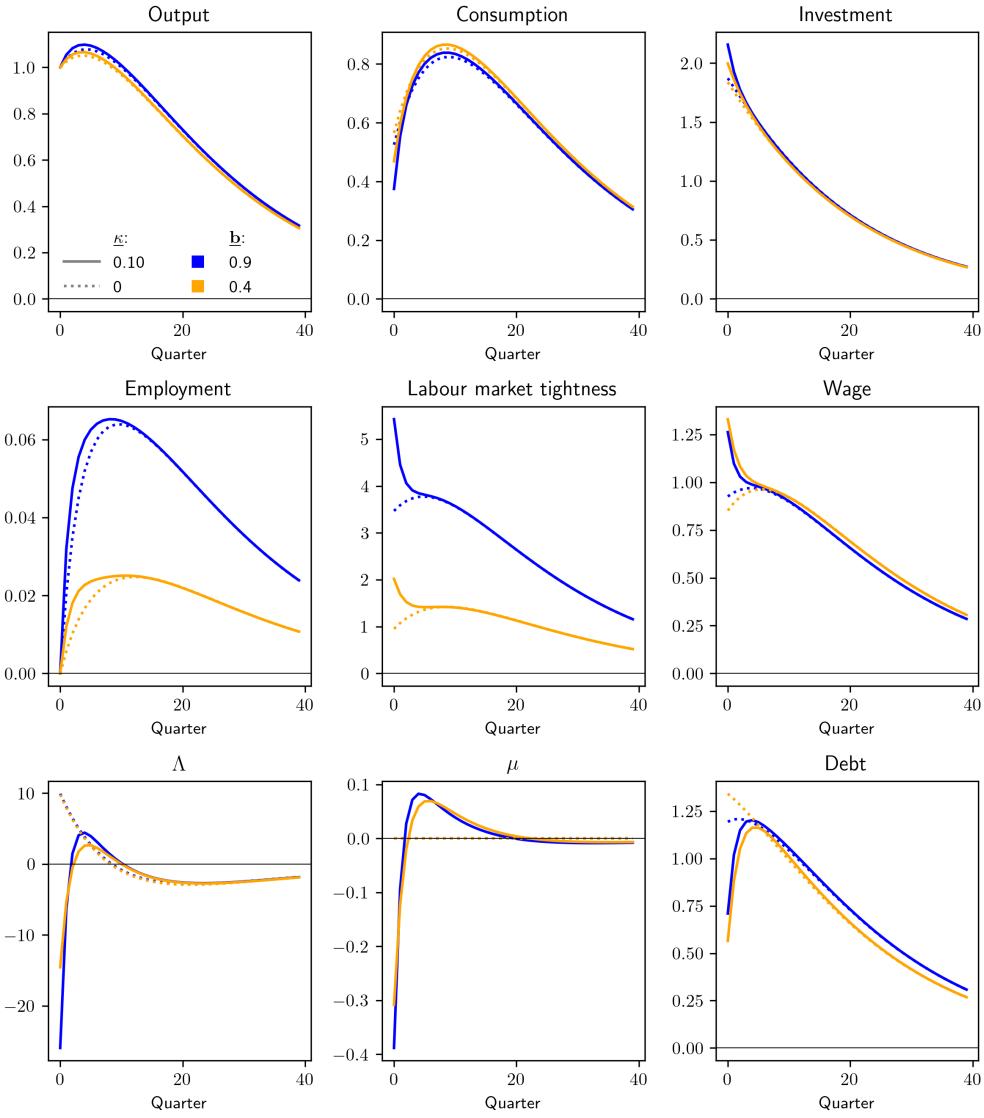


Figure 10: Low vs. High Fundamental Surplus: Positive 1% Productivity Shock. Percentage deviation from the steady state.

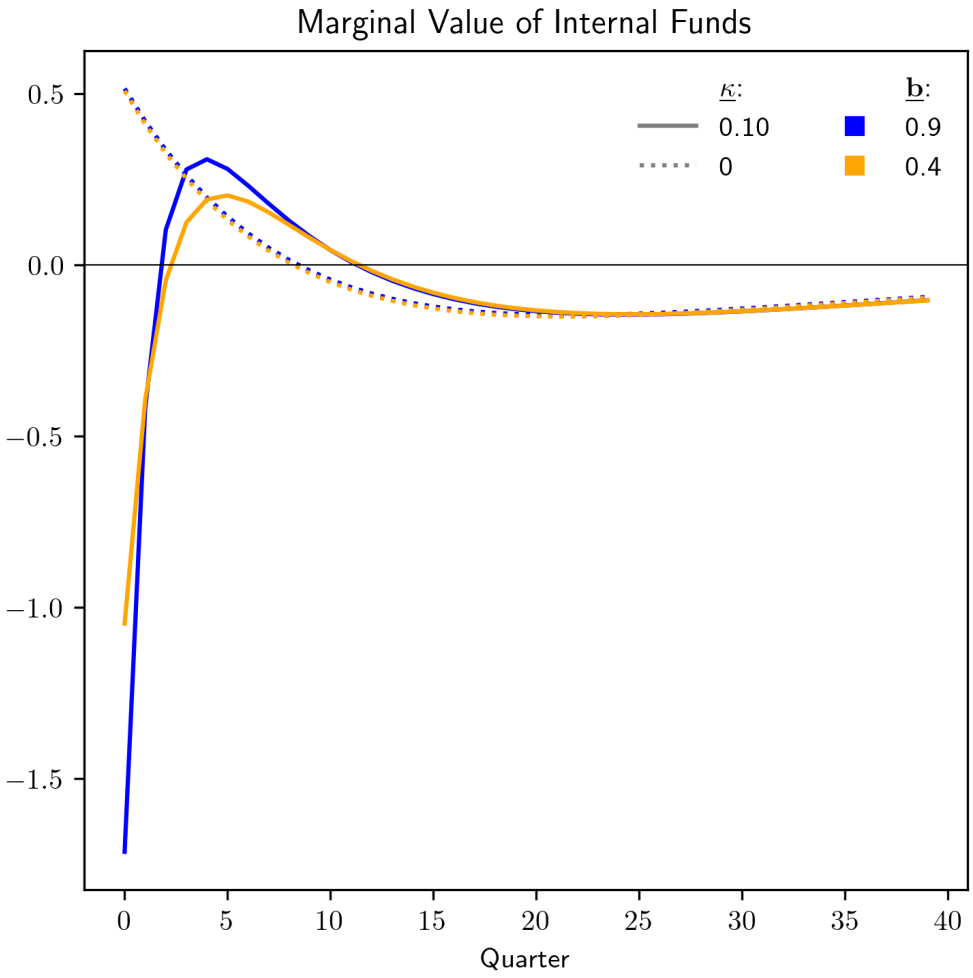


Figure 11: Low vs. High Fundamental Surplus: Positive 1% Productivity Shock. Percentage deviation from the steady state.

With dividend adjustment costs, however, the low fundamental surplus amplifies the response of debt.

6.2.2 Borrowing Constraint Shock

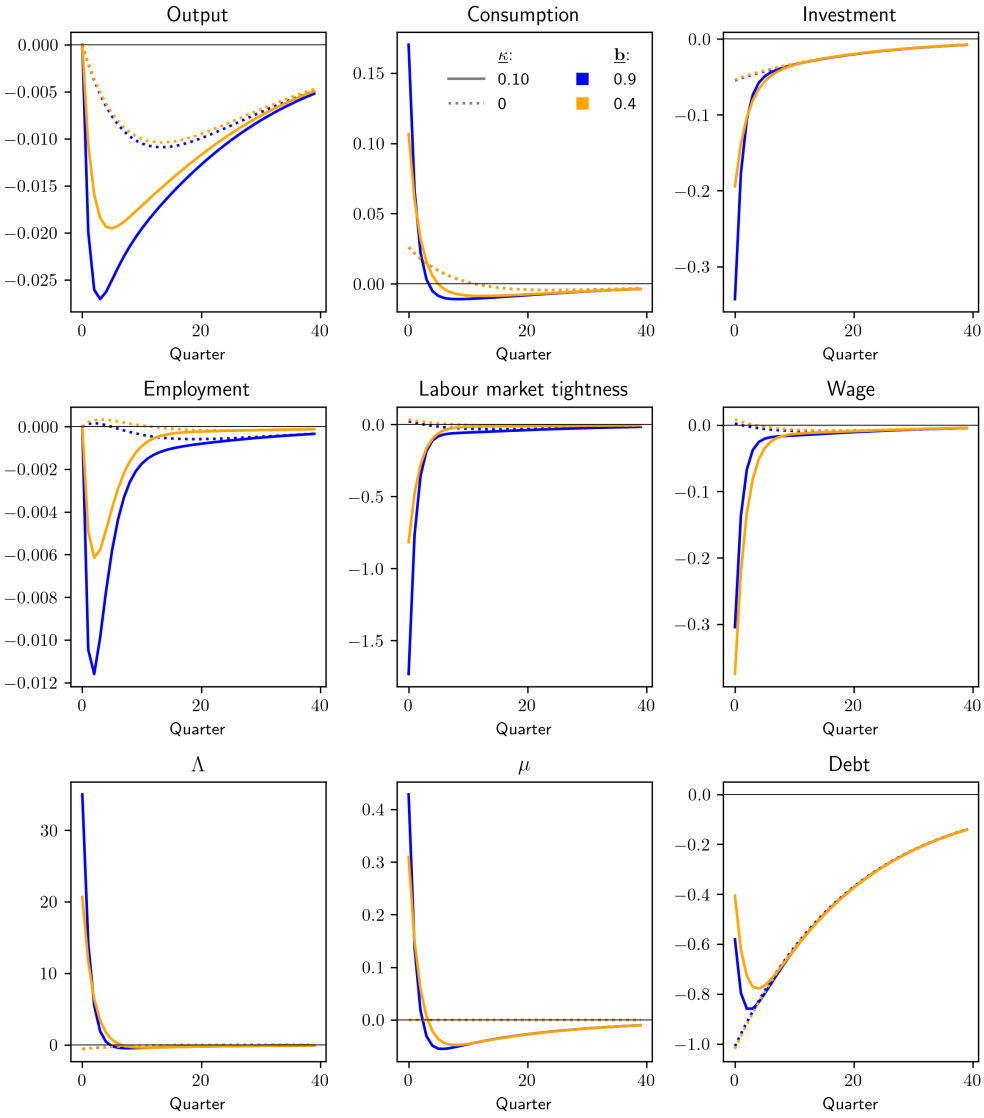


Figure 12: Low vs. High Fundamental Surplus: Negative 1% Borrowing Constraint Shock. Percentage deviation from the steady state.

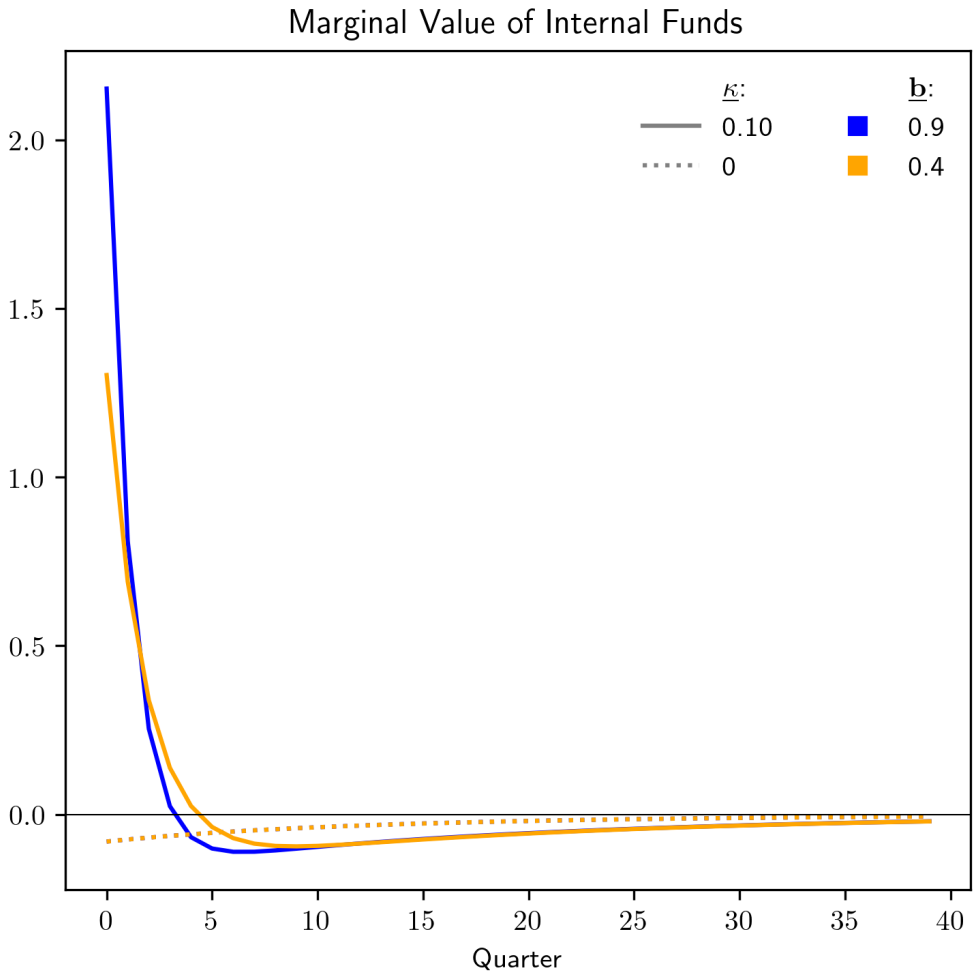


Figure 13: Low vs. High Fundamental Surplus: Negative 1% Borrowing Constraint Shock. Percentage deviation from the steady state.

Figure 12 repeats the experiment for a -1% borrowing-constraint shock and reports the same variables as before, while Figure 13 shows the marginal value of internal funds. The figure shows that, through the marginal value of internal funds, a low fundamental surplus amplifies responses in the labour, goods, and financial markets, particularly at or near the shock's impact. With no dividend adjustment costs and a weak marginal value of internal funds, however, the shock does not transmit into the labour market. With $\kappa > 0$, peak output is amplified by approximately 40% between the $b = 0.4$ and $b = 0.9$

economies, with much larger amplifications in labour-market tightness and borrowing-constraint tightness. Intuitively, the shock reduces the firm's borrowing capacity, and the firm responds by cutting vacancies because the rising marginal value of internal funds increases the marginal cost of posting vacancies. When the fundamental surplus is low, the firm already has fewer resources to post vacancies; therefore, to recover some borrowing capacity, they cut more vacancies. As a consequence, the workforce reduces by more, leading to a larger decline in investment and spilling over into lower cash flow.

7 Conclusion

This paper shows that interactions between typical labour and financial market frictions explain the close links between the two markets. To do this, the paper develops a simple model with DMP labour-search frictions, costly dividend adjustment, and borrowing constraints. The model shows that the two markets are linked through the firm's marginal value of internal funds. EBITDA borrowing constraints, together with dividend adjustment costs, induce sufficiently large changes in the firm's marginal value of internal funds, and bring the model closer to the labour-financial market relation observed in the data than an asset-based borrowing constraint. In addition, the model shows that, through the marginal value of internal funds, a low fundamental surplus amplifies transmission between labour and financial markets.

References

Andolfatto, D., 1996. Business cycles and labor-market search. *The american economic review*, pp.112–132.

Bell, B. and Machin, S., 2018. Minimum wages and firm value. *Journal of labor economics*, 36(1), pp.159–195.

Board of Governors of the Federal Reserve System, 2026. Net percentage of domestic banks tightening standards for commercial and industrial loans to large and middle-market firms. retrieved from FRED, Federal Reserve Bank of St. Louis. Series DRTSCILM, https://fred.stlouisfed.org/series/DRTSCILM.

Chodorow-Reich, G., 2014. The employment effects of credit market disruptions: Firm-level evidence from the 2008–9 financial crisis. *The quarterly journal of economics*, 129(1), pp.1–59.

Congressional Budget Office, 2017. *International comparisons of corporate income tax rates*. Washington, DC: Congressional Budget Office.

Drechsel, T., 2023. Earnings-based borrowing constraints and macroeconomic fluctuations. *American economic journal: Macroeconomics*, 15(2), pp.1–34.

Fujita, S. and Ramey, G., 2012. Exogenous versus endogenous separation. *American economic journal: Macroeconomics*, 4(4), pp.68–93.

Garin, J., 2015. Borrowing constraints, collateral fluctuations, and the labor market. *Journal of economic dynamics and control*, 57, pp.112–130.

Hagedorn, M. and Manovskii, I., 2008. The cyclical behavior of equilibrium unemployment and vacancies revisited. *American economic review*, 98(4), pp.1692–1706.

Hall, R.E., 2005. Employment fluctuations with equilibrium wage stickiness. *American economic review*, 95(1), pp.50–65.

Jermann, U. and Quadrini, V., 2012. Macroeconomic effects of financial shocks. *American economic review*, 102(1), pp.238–271.

Kiyotaki, N. and Moore, J., 1997. Credit cycles. *Journal of political economy*, 105(2), pp.211–248.

Kuehn, L.A., Simutin, M. and Wang, J.J., 2017. A labor capital asset pricing model. *The journal of finance*, 72(5), pp.2131–2178.

Lian, C. and Ma, Y., 2021. Anatomy of corporate borrowing constraints. *The quarterly journal of economics*, 136(1), pp.229–291.

Ljungqvist, L. and Sargent, T.J., 2017. The fundamental surplus. *American economic review*, 107(9), pp.2630–2665.

Merz, M., 1995. Search in the labor market and the real business cycle. *Journal of monetary economics*, 36(2), pp.269–300.

Moiseeva, Y., 2026. The interaction between credit and labor market frictions. *International economic review*, 67(4), pp.1309–1330.

Monacelli, T., Quadrini, V. and Trigari, A., 2023. Financial markets and unemployment. *Journal of financial economics*, 147(3), pp.596–626.

Rouwenhorst, K.G., 1995. Asset pricing implications of equilibrium business cycle models. *Frontiers of business cycle research*, 1, pp.294–330.

Shimer, R., 2004. The consequences of rigid wages in search models. *Journal of the european economic association*, 2(2-3), pp.469–479.

Shimer, R., 2005. The cyclical behavior of equilibrium unemployment and vacancies. *American economic review*, 95(1), pp.25–49.

U.S. Bureau of Labor Statistics, 2026a. Job openings: Total nonfarm. retrieved from FRED, Federal Reserve Bank of St. Louis. Series JTSJOL, https://fred.stlouisfed.org/series/JTSJOL.

U.S. Bureau of Labor Statistics, 2026b. Unemployment level. retrieved from FRED, Federal Reserve Bank of St. Louis. Series UNEMPLOY, https://fred.stlouisfed.org/series/UNEMPLOY.

A The Firm's Problem

Let us begin with the firm's optimisation problem,

$$\begin{aligned} J(\omega; S) = \max_{d,v,k',x'} \Bigg\{ & d + \mathbb{E}\left[m' J(\omega'; S')\right] \\ & + \mu \left[y - wn - \gamma v + \frac{x'}{\widetilde{R}} - x - k' + (1-\delta)k - \psi(d) \right] \\ & + \Lambda \left[\xi \left(y - wn - \gamma v \right) - \frac{x'}{\widetilde{R}} \right] \Bigg\} \end{aligned}$$

Dividend optimality condition The first-order condition with respect to d is

$$0 = 1 - \mu \psi'(d)$$

which rearranges to (11).

Vacancy optimality condition, and value of an additional worker Turning to the first order condition with respect to vacancies,

$$0 = \mathbb{E}\left[m' \frac{\partial J(\omega'; S')}{\partial v} \right] - \gamma\mu - \Lambda\xi\gamma$$

Since $\frac{\partial J(\omega';S')}{\partial v} = \frac{\partial J(\omega';S')}{\partial n'} \times \frac{\partial n'}{\partial v}$, and since $\partial n'/\partial v = q$, writing $J_n(\omega'; S') = \frac{\partial J(\omega';S')}{\partial n'}$, and knowing that $\mu = 1 + \hat{\mu}$ from the main paper, the above may be written as,

$$0 = q\mathbb{E}\left[m' J_n(\omega'; S') \right] - \gamma(1 + \hat{\mu}) - \Lambda\xi\gamma$$

therefore,

$$q\mathbb{E}\left[m'J_n(\omega';S')\right] = \gamma(1+\hat{\mu}+\Lambda\xi)$$

It also means that taking the envelope condition with respect to labour, and knowing from the law of motion of labour that $\partial n'/\partial n = 1-s$,

$$J_n(\omega;S) = (1+\hat{\mu})[zF_n(k,n)-w] + \xi\Lambda[zF_n(k,n)-w] + (1-s)\mathbb{E}\left[m'J_n(\omega';S')\right]$$

which can be written as,

$$J_n(\omega;S) = (1+\hat{\mu}+\xi\Lambda)[zF_n(k,n)-w] + \xi\Lambda[zF_n(k,n)-w] + (1-s)\mathbb{E}\left[m'J_n(\omega';S')\right]$$

This is the value of an additional worker in the main paper.

Capital optimality condition Turning to the first-order condition with respect to capital, k', letting $J_k(\omega;S) = \partial J(\omega;S)/\partial k$,

$$0 = \mathbb{E}\left[m'J_k(\omega';S')\right] - (1+\hat{\mu})$$

To determine $J_k(\omega';S')$, take the envelope condition with respect to k,

$$J_k(\omega;S) = (1+\hat{\mu})\left[zF_k(k,n)+(1-\delta)\right] + \xi\Lambda zF_k(k,n)$$

This means, we can write the new capital optimality condition as,

$$\mathbb{E}\left[m'\left\{(1+\hat{\mu}')\left[z'F_k(k',n')+(1-\delta)\right] + \xi'\Lambda'z'F_k(k',n')\right\}\right] = 1+\hat{\mu}$$

Capital - Labour ratio in the steady state In the steady state, $m = 1/R = \beta = (1+\rho)^{-1}$, and $\hat{\mu} = \hat{\mu}'$, which turns the capital optimality condition as,

$$\frac{1}{1+\rho}\left\{(1+\hat{\mu})[zF_k(k,n)+(1-\delta)]+\xi\Lambda zF_k(k,n)\right\} = 1+\hat{\mu}$$

This can we re-written as,

$$zF_k(k,n)+(1-\delta)+\frac{\xi\Lambda}{1+\hat{\mu}}zF_k(k,n) = 1+\rho$$

therefore,

$$zF_k(k,n)\left[1+\xi\frac{\Lambda}{1+\hat{\mu}}\right] = \rho+\delta$$

Since $zF_k(k,n) = z\alpha k^{\alpha-1}n^{1-\alpha}$, then,

$$z\alpha\left(\frac{k}{n}\right)^{\alpha-1}\left[1+\xi\frac{\Lambda}{1+\hat{\mu}}\right] = \rho+\delta$$

which can be re-arranged to,

$$\frac{k}{n} = \left[\frac{\alpha z\left(1+\xi\frac{\Lambda}{1+\hat{\mu}}\right)}{\rho+\delta}\right]^{\frac{1}{1-\alpha}}$$

Debt optimality condition The first order condition with respect to x' is,

$$0 = \mathbb{E}\left[m'\frac{\partial J(\omega';S')}{\partial x'}\right]+\frac{1+\hat{\mu}}{\tilde{R}}-\frac{\Lambda}{\tilde{R}}$$

Computing the envelope condition with respect to debt,

$$\frac{\partial J(\omega;S)}{\partial x} = -(1+\hat{\mu})$$

which means the optimality can be written as,

$$\frac{\Lambda}{\tilde{R}} + \mathbb{E}\left[m'(1+\hat{\mu}')\right] = \frac{1+\hat{\mu}}{\tilde{R}}$$

B Dynamic Nash Wage Derivation

Recall that the household's Bellman equation is,

$$W(a,n;S) = \max_{c,a'} \left\{u(c) + \beta \mathbb{E}\left[W(a',n';S')\right]\right\}$$

subject to (21) and (22). We can compute the envelope condition with respect to labour as,

$$\frac{\partial W(a,n;S)}{\partial n} = W_n(a,n;S) = u'(c)(w-b) + \beta \frac{\partial n'}{\partial n} \mathbb{E}\left[W_n(a',n';S')\right]$$

Since $\partial n'/\partial n = 1-s-f$, this means,

$$W_n(a,n;S) = u'(c)(w-b) + \beta(1-s-f)\mathbb{E}\left[W_n(a',n';S')\right] \tag{28}$$

Taking the first-order condition of (24) with respect to w gives us,

$$\frac{\Psi}{1-\Psi} \frac{J_n(\omega;S)}{1+\hat{\mu}+\xi\Lambda} = \frac{W_n(a,n;S)}{u'(c)} \tag{29}$$

where we divide the firm's side by $1+\hat{\mu}+\xi\Lambda$ because $\partial J_n/\partial w = -(1+\hat{\mu}+\xi\Lambda)$.

We can re-write (28) as,

$$\frac{W_n(a,n;S)}{u'(c)} = (w-b) + \beta(1-s-f)\mathbb{E}\left[\frac{u'(c')}{u'(c)}\frac{W_n(a',n';S')}{u'(c')}\right]$$

Since from the Euler equation $u'(c) = \beta R\mathbb{E}[u'(c')]$,

$$\frac{W_n(a,n;S)}{u'(c)} = (w-b) + \frac{1}{R}(1-s-f)\mathbb{E}\left[\frac{W_n(a',n';S')}{u'(c')}\right]$$

Which we can rewrite as,

$$\frac{\Psi}{1-\Psi}\frac{J_n(\omega;S)}{1+\hat{\mu}+\xi\Lambda} = (w-b) + \frac{1}{R}(1-s-f)\frac{\Psi}{1-\Psi}\mathbb{E}\left[\frac{J_n(\omega',S')}{1+\hat{\mu}'+\xi'\Lambda'}\right] \tag{30}$$

Now we can take (16), which recall is,

$$J_n(\omega;S) = (1+\hat{\mu}+\xi\Lambda)\left[zF_n(k,n)-w\right] + (1-s)\mathbb{E}\left[m'J_n(\omega';S')\right]$$

and the free-entry condition (15), which is,

$$q\mathbb{E}\left[m'J_n(\omega',S')\right] = \gamma(1+\hat{\mu}+\xi\Lambda)$$

which, since $m' = \beta u'(c')/u'(c) = 1/R$, lets us rewrite the value of an additional worker as,

$$J_n(\omega;S) = (1+\hat{\mu}+\xi\Lambda)\left[zF_n(k,n)-w\right] + (1-s)\frac{\gamma(1+\hat{\mu}+\xi\Lambda)}{q} \tag{31}$$

Substituting (31) and (15) into (30),

$$\frac{\Psi}{1-\Psi}\frac{1}{(1+\hat{\mu}+\xi\Lambda)}\left[(1+\hat{\mu}+\xi\Lambda)\left[zF_n(k,n)-w\right] + (1-s)\frac{\gamma(1+\hat{\mu}+\xi\Lambda)}{q}\right] = (w-b)+(1-s-f)\frac{\Psi}{1-\Psi}\mathbb{E}\left[\frac{m'J_n(\omega';S')}{(1+\hat{\mu}'+\xi'\Lambda')}\right]$$

Multiplying the LHS and RHS by $1-\Psi$,

$$\Psi z F_n(k,n) - \Psi w + \frac{\Psi(1-s)\gamma}{q} = w - \Psi w - (1-\Psi)b + (1-s-f)\Psi\mathbb{E}\left[\frac{m' J_n(\omega';S')}{1+\hat{\mu}'+\xi'\Lambda'}\right]$$

making the wage the subject,

$$w = \Psi\left[zF_n(k,n) + \frac{(1-s)\gamma}{q}\right] + (1-\Psi)b - (1-s-f)\Psi\mathbb{E}\left[\frac{m' J_n(\omega';S')}{1+\hat{\mu}'+\xi'\Lambda'}\right] \tag{32}$$

Since $q\mathbb{E}\left[m' J_n(\omega';S')\right] = \gamma(1+\hat{\mu}+\xi\Lambda)$, then, the above becomes,

$$w = \Psi\left[zF_n(k,n) + \frac{(1-s)\gamma}{q}\right] + (1-\Psi)b - (1-s-f)\Psi\frac{\gamma}{q}\mathbb{E}\left[\frac{1+\hat{\mu}+\xi\Lambda}{1+\hat{\mu}'+\xi'\Lambda'}\right]$$

this can be simplified to,

$$w = \Psi\left[zF_n(k,n) + \frac{(1-s)\gamma}{q} - (1-s-f)\frac{\gamma}{q}\mathbb{E}\left[\frac{1+\hat{\mu}+\xi\Lambda}{1+\hat{\mu}'+\xi'\Lambda'}\right]\right] + (1-\Psi)b$$

which can be written as,

$$w = \Psi\left[zF_n(k,n) + \frac{(1-s)\gamma}{q} - (1-s)\frac{\gamma}{q}\mathbb{E}\left[\frac{1+\hat{\mu}+\xi\Lambda}{1+\hat{\mu}'+\xi'\Lambda'}\right] + f\frac{\gamma}{q}\mathbb{E}\left[\frac{1+\hat{\mu}+\xi\Lambda}{1+\hat{\mu}'+\xi'\Lambda'}\right]\right] + (1-\Psi)b$$

Factorising $(1-s)\gamma/q$ and knowing that $\theta = f/q$ allows us to obtain,

$$w = \Psi\left[zF_n(k,n) + \frac{(1-s)\gamma}{q}\left[1 - \mathbb{E}\left[\frac{1+\hat{\mu}+\xi\Lambda}{1+\hat{\mu}'+\xi'\Lambda'}\right]\right] + \gamma\theta\mathbb{E}\left[\frac{1+\hat{\mu}+\xi\Lambda}{1+\hat{\mu}'+\xi'\Lambda'}\right]\right] + (1-\Psi)b$$

which is the dynamic Nash wage.

C Nash wage in the stationary equilibrium

In the steady state $\mathbb{E}\left[\frac{1+\hat{\mu}+\xi\Lambda}{1+\hat{\mu}'+\xi'\Lambda'}\right] = 1$, then,

$$w = \Psi\left[zF_n(k,n) + \gamma\theta\right] + (1-\Psi)b$$

D Deriving the Fundamental Surplus

In the steady state, (31) and the free-entry condition (15) give $q\beta J_n = \gamma(1+\hat{\mu}+\xi\Lambda)$ with $J_n = (1+\hat{\mu}+\xi\Lambda)[zF_n(k,n) - w]/(1-\beta(1-s))$. The factor $(1+\hat{\mu}+\xi\Lambda)$ appears on both sides and cancels, leaving the free-entry condition of the frictionless economy,

$$\frac{(r+s)\gamma}{q} = zF_n(k,n) - w$$

An analogue to the steady state Nash wage,

$$w = b + \frac{\Psi}{1-\Psi}(r+s+f)\left(\frac{\gamma}{q}\right)$$

Combining the two and rearranging gives us,

$$\frac{(r+s)\gamma}{q} = zF_n(k,n) - b - \frac{\Psi}{1-\Psi}(r+s+f)\frac{\gamma}{q}$$

which, after rearranging, gives

$$FS = zF_n(k,n) - b = \frac{\gamma}{q}\left[r+s+\frac{\Psi}{1-\Psi}(r+s+f)\right]$$

which is the fundamental surplus.

To compute the elasticity of labour market tightness to productivity we differentiate

$$FS = zF_n - b = \frac{\gamma}{q(\theta)}\left[r + s + \frac{\Psi}{1-\Psi}(r + s + \theta q(\theta))\right]$$

with respect to z, knowing that $q(\theta) = \Upsilon\theta^{-\eta}$, means that at the steady state,

$$\eta_{\theta,z} = \frac{r + s + \Psi\theta q}{\eta(r + s) + \Psi\theta q} \cdot \frac{zF_n(k, n)}{zF_n(k, n) - b}$$

The second factor is the Ljungqvist and Sargent (2017) inverse-fundamental-surplus term.

E A Frictionless RBC–DMP Baseline

This section sets out the frictionless baseline economy.

Households The household's problem remains the same as before, but there is no tax. This means that the household's budget constraint is,

$$c + \frac{a'}{R} = wn + (1 - n)b + a + d$$

and the Euler equation remains the same, $u'(c) = R\beta\mathbb{E}[u'(c')]$.

Firms Firms do not confront a borrowing constraint. The firm solves,

$$J(\omega;S) = \max_{d,v,k',x'} \Bigg\{ d + \mathbb{E}\left[m' J(\omega';S')\right] \\ + \mu \left[y - wn - \gamma v + \frac{x'}{R} - x - k' + (1-\delta)k - \psi(d) \right] \Bigg\}$$

The optimality conditions yield, where $\mu = 1 + \hat{\mu}$ as before. With regards to dividends,

$$1 = (1+\hat{\mu})\psi'(d)$$

with respect to vacancies,

$$q\,\mathbb{E}[m' J_n(\omega';S')] = \gamma(1+\hat{\mu})$$

and the envelope condition with respect to labour is,

$$J_n(\omega;S) = (1+\hat{\mu})\left[zF_n(k,n) - w\right] + (1-s)\mathbb{E}\left[m' J_n(\omega';S')\right]$$

with respect to capital,

$$\mathbb{E}\left[m'\left\{(1+\hat{\mu}')\left(z'F_k(k',n') + (1-\delta)\right)\right\}\right] = 1 + \hat{\mu}$$

with respect to debt,

$$\mathbb{E}\left[m'(1+\hat{\mu}')\right] = \frac{1+\hat{\mu}}{R}$$

Nash wages With dividend adjustment costs, the Nash wage becomes,

$$w^{\text{Nash}} = \Psi \left[zF_n(k,n) + \frac{(1-s)\gamma}{q} - (1-s-f)\frac{\gamma}{q}\mathbb{E}\left[\frac{1+\hat{\mu}}{1+\hat{\mu}'}\right]\right] + (1-\Psi)b$$

Recursive Competitive Equilibrium The recursive competitive equilibrium in this model is defined similarly to the one in the main body. To close the model, $X' = K'$ is imposed to pin down the quantity of debt, thus ensuring that the aggregate resource constraint $Y(S) + (1-N)b - \gamma V(S) = C(S) + K'(S) - (1-\delta)K + \kappa(D(S)-\phi)^2$ holds.

F Kiyotaki-Moore Borrowing Constrained Economy

This section presents the Kiyotaki-Moore borrowing constrained economy.

Households. The households set up remains the same as in the main paper.

Firms. Firms confront the borrowing constraint,

$$\frac{x'}{\widetilde{R}} \leq \xi^{KM}(1-\delta)k'.$$

and solve the problem,

$$\begin{aligned} J(\omega;S) = \max_{d,v,k',x'} \Bigg\{ & d + \mathbb{E}\left[m' J(\omega';S')\right] \\ & + \mu\left[y - wn - \gamma v + \frac{x'}{\widetilde{R}} - x - k' + (1-\delta)k - \psi(d)\right] \\ & + \Lambda\left[\xi^{KM}(1-\delta)k' - \frac{x'}{\widetilde{R}}\right] \Bigg\} \end{aligned}$$

Using the same notation that $\mu \equiv 1 + \hat{\mu}$, we can write the first order condition with respect to dividends as,

$$1 = (1 + \hat{\mu})\psi'(d)$$

The optimality condition with respect to vacancies is,

$$q\mathbb{E}\left[m' J_n(\omega'; S')\right] = \gamma(1 + \hat{\mu})$$

where the value of an additional worker is,

$$J_n(\omega; S) = (1 + \hat{\mu})\left[zF_n(k, n) - w\right] + (1 - s)\mathbb{E}\left[m' J_n(\omega'; S')\right]$$

Capital investment's optimality condition is,

$$\mathbb{E}\left[m' J_k(\omega'; S')\right] + \Lambda\xi^{KM}(1 - \delta) = 1 + \hat{\mu}$$

where capital's envelope condition is,

$$J_k(\omega; S) = (1 + \hat{\mu})\left[zF_k(k, n) + 1 - \delta\right]$$

Combining the two yields,

$$\mathbb{E}\left[m'\Big\{(1 + \hat{\mu}')\left[z' F_k(k', n') + 1 - \delta\right]\Big\}\right] + \Lambda\xi^{KM}(1 - \delta) = 1 + \hat{\mu}$$

Debt's optimality condition can be computed by combining the first order condition,

$$\mathbb{E}\left[m' \frac{\partial J(\omega'; S')}{\partial x'}\right] + \frac{1 + \hat{\mu}}{\widetilde{R}} = \frac{\Lambda}{\widetilde{R}}$$

and the envelope condition,

$$\frac{\partial J(\omega; S)}{\partial x} = -(1 + \hat{\mu})$$

therefore,

$$\frac{\Lambda}{\widetilde{R}} + \mathbb{E}\left[m'(1 + \hat{\mu}')\right] = \frac{1 + \hat{\mu}}{\widetilde{R}}$$

Nash Wage The Nash wage is the same as in the baseline with dividend adjustment costs,

$$w^{\text{Nash}} = \Psi \left[zF_n(k, n) + \frac{(1 - s)\gamma}{q} - (1 - s - f)\frac{\gamma}{q}\mathbb{E}\left[\frac{1 + \hat{\mu}}{1 + \hat{\mu}'}\right]\right] + (1 - \Psi)b$$

Recursive Competitive Equilibrium The recursive competitive equilibrium is defined in a similar manner as in the main text.